

# KMT-2026-BLG-0521 / OGLE-2026-BLG-0058

## The restricted-set reductions (v4 and v5) — standalone report

Prepared 2026-09-08. Solutions: `~/KMTdata/Results/v4b/IFfinal_<field>.mat` and `~/KMTdata/Results/v5/IFfinal_<field>.mat` (variable IFsys in each). This document describes **only** these two runs, which share one source set and differ in exactly one respect: v4 pins the frame to Gaia during the fit, v5 fits everything freely and ties to Gaia afterwards. The decade reduction that remains the project's reference solution is described in `KMT260058_astrometry_report.pdf`.

### 1. What this run is

A reduction built to a different specification from the reference one, to test two ideas: restrict the analysis to a clean calibration set, and tie the frame to Gaia from the start rather than afterwards.

1. **Analysis set.**  $14 < I_{\text{OGLE}} < 17$ , no companion brighter than  $I = 18$  within 2.0 arcsec, and a Gaia counterpart with **RUWE < 1.4**. The target is added explicitly — at  $I = 18.13$  it falls outside the magnitude window. Every other source is **dropped at the input**, so all measurements, detrending and quality cuts act on this set alone. 2. **The frame is pinned to Gaia.** Each calibration star's position *and* proper motion are held at Gaia's values, converted into cut-out pixels. The per-epoch affine is therefore fitted against Gaia positions propagated to that epoch by Gaia's own motions. The target is left free; pinning it would erase the signal. The six-parameter gauge is gone by construction and no post-hoc tie is needed. 3. **Passengers.** Three stars at the target's own magnitude ( $I \sim 18$ ), outside the calibration set, are carried along unpinned so that something comparable to the target can be plotted in the same frame.

|                                                                            | BLG41      | BLG01      |
|----------------------------------------------------------------------------|------------|------------|
| raw sources in the MSc file                                                | 1177       | 1187       |
| surviving the standard quality cuts — <b>the reference reduction's set</b> | <b>594</b> | <b>621</b> |
| <b>analysis set of these runs</b>                                          | <b>130</b> | <b>100</b> |
| of which pinned to Gaia (v4 only)                                          | 126        | 96         |
| free: target + passengers                                                  | 4          | 4          |
| epochs                                                                     | 19 133     | 19 981     |

The row that matters for comparison is the second: the reference reduction fits **594 and 621** sources, and these runs fit 130 and 100 of them. The raw file holds about 1180 entries, but half of those are marginal detections that the standard conversion discards before any reduction sees them — they are found in a median of **10%** of epochs, against 80% for the sources that survive. Every source detected in more than half the epochs is kept, without exception. So the restriction imposed here is 130 out of 594, not 130 out of 1177.

### 2. The calibration set

Selection, all conditions required:

- $14 < I_{\text{OGLE}} < 17$

- **no companion brighter than  $I = 18$  within 2.0 arcsec** (5 KMT pixels at 0.4"/pix), the companion list being the OGLE catalogue, which resolves about six times more sources than KMT detects
- **a Gaia DR3 counterpart with RUWE < 1.4**

|                                   | BLG41      | BLG01     |
|-----------------------------------|------------|-----------|
| in the magnitude window, isolated | 144        | 127       |
| with a Gaia counterpart           | 144        | 116       |
| <b>after RUWE &lt; 1.4</b>        | <b>126</b> | <b>96</b> |

The RUWE cut removes 12% and 17% — consistent with the 87% pass rate for  $G < 17$  across the whole cone, so crowding is not biasing it heavily.

### 3. Gaia

**RUWE is not in the local catsHTM DR3 table.** It carries `astrometric_chi2_al` and `astrometric_n_good_obs_al` but neither RUWE nor the `u0(G, BP-RP)` table needed to normalise UWE into it. The true values were queried from the Gaia archive instead — 9464 sources in a 3 arcmin cone, cached in `~/KMTdata/GaiaRef/`. This mattered: at  $G < 17$ , **87%** pass RUWE < 1.4, where the unnormalised UWE proxy predicted 74%, so the proxy would have discarded about 13% of good stars for no reason.

Gaia's proper-motion errors for these stars are **0.070 / 0.044 mas/yr**, some fifty times better than our internal frame. That is what makes pinning worthwhile in principle.

**The Gaia-to-pixel map is fitted directly** from the matched pairs, not carried through OGLE. The OGLE bridge is needed only to identify which star is which; routing coordinates through it inherits OGLE's ~110 mas astrometric error.

|                                | through OGLE  | fitted directly    |
|--------------------------------|---------------|--------------------|
| BLG41, median offset / scatter | 252 / 159 mas | <b>23 / 16 mas</b> |
| BLG01                          | 351 / 148 mas | <b>21 / 20 mas</b> |

The fitted scale is 2.519 (BLG41) and 2.529 (BLG01) pix/arcsec against the 2.500 expected at 0.4"/pix — an independent check that the map is right. The remaining ~20 mas is static per star, so it displaces each star's residuals by a constant and does not affect proper motion or any time-dependent signal.

### 4. The chain

With proper motions supplied by Gaia, the joint proper-motion fit and the per-season fits of the reference reduction are unnecessary. Two steps remain:

1. **Full-decade SysRem correction**, frame pinned, 2 components on the astrometric residuals. 2. **Final solution** reusing that correction, with the pixel-phase term on, 15 weighted iterations before SysRem and 6 after.

Convergence over the frame stars, last three iterations: **0.0009 mas (BLG41)** and **0.0026 mas (BLG01)**.

## 5. Results

|                                                | BLG41             | BLG01             |
|------------------------------------------------|-------------------|-------------------|
| frame-star residual RMS, $\Delta X / \Delta Y$ | 6.13 / 5.20 mas   | 5.30 / 5.42 mas   |
| target residual RMS, $\Delta X / \Delta Y$     | 29.62 / 22.63 mas | 28.73 / 23.25 mas |

For comparison, the reference reduction gives bright-star 6.41 / 6.88 and 6.48 / 7.01, and target **24.62 / 21.00** and **25.48 / 21.78**.

**The calibration stars are measured better and the target worse.** The stars are bright and isolated, which is what they were selected for; the target is **14 to 20% worse** than in the reference reduction. The cause is the frame: it is now built from 126 stars rather than ~600 weighted sources, which roughly doubles its ideal noise and raises its leverage by a factor of twenty to fifty. Restricting the analysis set is what costs. The Gaia tie is not at fault.

## 6. Proper motions, and whether the two fields agree

Converting the target's motion out of pixel axes with each field's own map:

|                         | BLG41           | BLG01           | difference           |
|-------------------------|-----------------|-----------------|----------------------|
| <b>v4, Gaia-pinned</b>  | -1.799 / -6.780 | -1.158 / -7.147 | <b>0.640 / 0.367</b> |
| reference, post-hoc tie | -2.92 / -7.25   | -2.22 / -7.24   | 0.702 / 0.004        |

mas/yr, as ( $\mu_\alpha \cos \delta$ ,  $\mu_\delta$ ). Both agree at about the 1 mas/yr level, which is the target's own uncertainty, but **pinning the frame did not tighten the agreement**. The post-hoc tie was already doing this job.

**The passengers agree better than the target does.** Three stars at the target's magnitude, freely fitted, carried through both fields:

| star                                  | BLG41         | BLG01          | difference         |
|---------------------------------------|---------------|----------------|--------------------|
| $l = 18.14$ , 4.5 pix from the target | -0.63 / -5.47 | -1.08 / -5.47  | 0.45 / 0.00        |
| $l = 17.91$ , 21 pix                  | -6.81 / -9.99 | -6.94 / -10.06 | <b>0.13 / 0.07</b> |
| $l = 18.05$ , 48 pix                  | +3.06 / -6.34 | +2.80 / -6.48  | <b>0.26 / 0.14</b> |
| <b>the target</b> , $l = 18.13$       | -1.80 / -6.78 | -1.16 / -7.15  | 0.64 / 0.37        |

The two clean passengers reproduce two to five times better than the target. The 4.5-pix star's apparent agreement is not a good sign — see section 7.

**Sensitivity to the set.** Adding only those three passengers to a set of 127 moved the fitted target motion by **0.42 / 0.63 mas/yr on BLG41** and 0.09 / 0.41 on BLG01. A result that moves by half a mas per year when three faint stars join the set is not a precise result, and this is the most direct measurement here of how uncertain the target's motion actually is.

## 7. A blended neighbour tracks the event

The passenger 4.5 pixels from the target behaves unlike anything else in the run. Comparing the 2026 season against the 2016–2022 baseline:

| star               | distance       | $\Delta X$ (BLG01 / BLG41) | $\Delta Y$ (BLG01 / BLG41) | season scatter |
|--------------------|----------------|----------------------------|----------------------------|----------------|
| target             | 0              | +2.7 / +2.6                | +7.1 / +6.4                | 3.6–5.3        |
| <b>passenger 1</b> | <b>4.5 pix</b> | <b>–44.1 / –57.8</b>       | <b>–27.9 / –32.2</b>       | 16–19          |
| passenger 2        | 21 pix         | +5.6 / +3.5                | +2.9 / +3.3                | 2.3–3.7        |
| passenger 3        | 48 pix         | –3.4 / –5.5                | –2.5 / –2.1                | 2.1–5.3        |

mas. The excursion is **40 to 58 mas at 4.5 pixels and gone by 21 pixels**, reproduced independently in both fields, and that star's season-to-season scatter is four times the others' in every season, not only in 2026. With a seeing FWHM near 7 pixels, a star 4.5 pixels away shares most of its light with the target: as the target brightens through the event, the blend's centroid moves. The neighbour moves *away* from the target, consistently in both fields, which is what PSF fitting a blend tends to do as flux is reattributed to the brightening component.

Three consequences:

- **That star is unusable as a comparison.** It was chosen as the closest at the target's magnitude, and being closest is exactly what disqualifies it. Passengers 2 and 3 are the meaningful ones.
- **Its good inter-field agreement in section 6 is misleading.** Both fields see the same blending artefact from the same event, so agreeing is not evidence of a good measurement.
- **It bears directly on the target.** The blend is mutual. If the neighbour's centroid moves 40 to 58 mas when the target brightens, the target's centroid is being pulled by the same shared flux. The target's own +2.7 / +7.1 mas in 2026 therefore cannot be read cleanly as astrometric microlensing; some or all of it may be blending. That is a systematic at the tens-of-mas level against a predicted signal of a few mas.

## 8. Overlap of the two fields

Matching the two source lists directly, after fitting the offset between the cut-outs (–1.11 / +0.24 pix):

| match radius   | BLG41 matched          | BLG01 matched          |
|----------------|------------------------|------------------------|
| 0.5 pix        | 339 / 594 (57%)        | 352 / 621 (57%)        |
| <b>1.0 pix</b> | <b>546 / 594 (92%)</b> | <b>575 / 621 (93%)</b> |
| 1.5 pix        | 576 / 594 (97%)        | 609 / 621 (98%)        |

These are indeed two observations of nearly the same sky, and the cut-outs span the same pixel range. The unmatched 7% are faint and marginal: median MAG\_PSF 17.18 against 16.87 for matched stars, and a detection fraction of 0.54 against 0.84. They are stars one reduction found and the other did not, or blends split differently. Restricted to MAG\_PSF < 17 the agreement is 94%.

Only 57% agree within 0.5 pix, so two independent reductions place a common star half a pixel to a pixel apart; 1 pixel is the appropriate radius.

Matching *through* Gaia and OGLE instead gives a misleading 58%, because their identification coverage over our sources is only 90% / 72% (Gaia) and 91% / 77% (OGLE) — sources neither catalogue identifies cannot be paired at all and count as non-overlapping.



## 9. v5 — the same set, freely fitted

v4 changes two things at once against the reference reduction: it shrinks the source set, and it pins the frame. v5 separates them. It uses **exactly the same source list** as v4b — read from the same file, not re-selected — the same two steps, the same iteration counts and the same SysRem and pixel-phase settings. The only difference is that nothing is pinned: every calibration star is freely fitted and takes part in the frame determination at every step, and Gaia enters only afterwards, as a post-hoc tie.

So **v3 to v5 measures what restricting the set costs**, and **v5 to v4b measures what pinning does**.

### Target residual RMS

|                                                 |  | BLG41 $\Delta X / \Delta Y$ | BLG01 $\Delta X / \Delta Y$ |
|-------------------------------------------------|--|-----------------------------|-----------------------------|
| <b>v3</b> — ~600 sources, free, post-hoc tie    |  | <b>24.62 / 21.00</b>        | <b>25.48 / 21.78</b>        |
| <b>v5</b> — 130/100 sources, free, post-hoc tie |  | 25.61 / 22.27               | 28.45 / 23.15               |
| <b>v4b</b> — the same sources, pinned to Gaia   |  | 29.62 / 22.63               | 28.73 / 23.25               |

| step     | isolates            | BLG41 X / Y    | BLG01 X / Y    |
|----------|---------------------|----------------|----------------|
| v3 → v5  | restricting the set | +4.0% / +6.0%  | +11.7% / +6.3% |
| v5 → v4b | pinning the frame   | +15.7% / +1.6% | +1.0% / +0.4%  |

Both hurt and neither dominates consistently. Restricting the set costs 4 to 12% everywhere; pinning costs almost nothing except in BLG41's X, where it costs 16%. The reference reduction remains the best for the target on every axis in both fields.

**Frame stars.** Over exactly the same 126 / 96 stars:

|                     | BLG41              | BLG01              |
|---------------------|--------------------|--------------------|
| v5, freely fitted   | <b>4.87 / 4.81</b> | <b>4.77 / 4.77</b> |
| v4b, pinned to Gaia | 6.13 / 5.20        | 5.30 / 5.42        |

4.8 mas is the best frame-star astrometry this project has produced. The comparison is not entirely fair — the free fit adjusts four parameters per star and so absorbs part of each star's own systematics, which the pinned version cannot — but the ~1.3 mas gap is the price of holding stars at catalogue values.

Convergence over the frame stars: 0.0041 mas (BLG41) and 0.0002 (BLG01).

### Inter-field agreement of the target's absolute motion.

|                          | difference between the fields |
|--------------------------|-------------------------------|
| v3                       | 0.702 / 0.004                 |
| v5, <code>gaiaTie</code> | 0.072 / 0.123                 |
| v5, positional-map tie   | 0.363 / 0.294                 |
| v4b                      | 0.640 / 0.367                 |

mas/yr. v5 agrees best, but **the headline 0.072 should not be quoted alone**. `gaiaTie` fits a 2 by 2 matrix from a proper-motion regression, and on these small sets it returned rotations of **146.34 deg for BLG41 and 108.65 for BLG01** — 38 deg apart, where the direct positional map puts the two cut-outs 0.126 deg apart. That matrix is fitting

noise. Re-tying independently, with the linear part taken from the positional map and only the gauge constant from the proper-motion comparison, gives 0.363 / 0.294 instead. The honest statement is that v5's fields agree to **0.07 to 0.36 in RA and 0.12 to 0.29 in Dec**, and that the choice of tie method is itself a systematic of 0.2 to 0.6 mas/yr.

The gauge removed from v5 is -0.219 / +4.758 mas/yr (BLG41, 130 calibrators) and +2.834 / +3.983 (BLG01, 100). With it removed, the seven plotted stars agree between the fields to 0.25 to 0.78 mas/yr in RA and 0.02 to 0.87 in Dec.

10. Verdict

Neither v4 nor v5 is adopted as the primary solution. Restricting the analysis to the calibration set costs 4 to 12% on the target, and pinning the frame to Gaia costs a further 0 to 16%; the reference reduction is better on every axis in both fields. What the pair does establish, by differencing, is that **both choices cost something and the set restriction is the more consistent penalty** — pinning is nearly free except on BLG41's X axis.

v5 nevertheless produces the best frame stars (4.8 mas) and the best inter-field agreement of the target's absolute motion of any reduction here, and it does so while leaving every calibration star free to take part in the frame. If a restricted-set reduction is wanted, v5 rather than v4 is the one to use.

Its most valuable output is not the astrometry but section 7: the direct measurement of how far a blended neighbour is dragged by the event, which applies to the target in every reduction we have made.

11. Figures and files

| file                                         | content                                                                                        |
|----------------------------------------------|------------------------------------------------------------------------------------------------|
| report_v4_RMS.png                            | residual RMS against OGLE I, per axis and field, for this run                                  |
| report_v4_motion_<field>_target.png          | the target, sidereal-month bins, 2x2                                                           |
| report_v4_motion_<field>_cal_d{24,30,43}.png | three calibration stars, proper motion <b>pinned to Gaia</b>                                   |
| report_v4_motion_<field>_out_d{04,21,48}.png | the three I ~ 18 passengers, freely fitted                                                     |
| report_v5_motion_<field>_*.png               | the same seven stars in <b>v5</b> , where nothing is pinned; the quoted sky motion is absolute |

The `_d NN` suffix is the star's distance from the target in pixels, and names the **same physical star in both fields** — the figures are directly comparable across fields. (An earlier version ranked the comparison stars per field, so `cal1` denoted different objects in BLG41 and BLG01.)

**Axis convention.** The Gaia-to-pixel map has a negative determinant: it contains a parity flip, and the pixel X axis runs *opposite* to increasing RA.

|       | rotation, flip removed | scale             |
|-------|------------------------|-------------------|
| BLG41 | -0.317 deg             | 2.5194 pix/arcsec |
| BLG01 | -0.444 deg             | 2.5290 pix/arcsec |

So the cut-outs are essentially aligned with RA and Dec apart from that X flip, with a third of a degree of residual rotation, and the two fields differ in orientation by only 0.126 deg. In each figure the **legend** quotes the motion along the

pixel axis and the **panel heading** quotes it on the sky, so the two carry opposite signs in X by construction; both are labelled accordingly.

In every motion figure the columns are the two axes, X left and Y right; the top row keeps the proper motion and the bottom row removes it. **A pinned star's red line is Gaia's proper motion, not a fitted one**, so its lower panels test the frame, whereas the target's and the passengers' test the fit.

## Figures

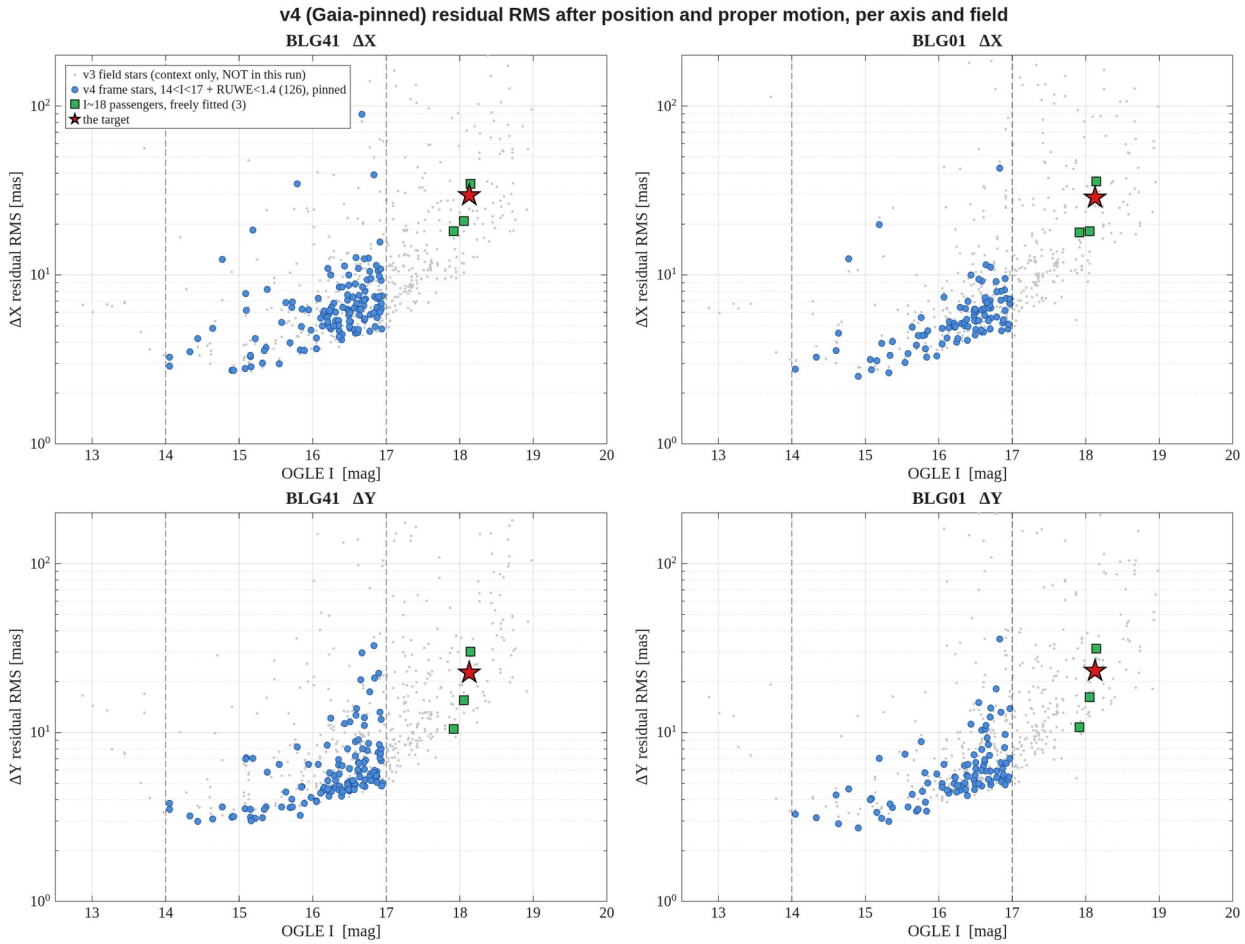


Figure 1. Residual RMS against the OGLE catalogue magnitude, per axis and per field, **for this run**, after position and proper motion have been removed. Blue: the 126 / 96 frame stars,  $14 < I < 17$  with  $RUWE < 1.4$ , pinned to Gaia. Green: the three 1~18 passengers, freely fitted. Red: the target. Grey: field stars from the reference reduction, shown for context only — **they are not part of this run**. Dashed lines mark the 14 and 17 mag limits.

report\_v4\_RMS.png

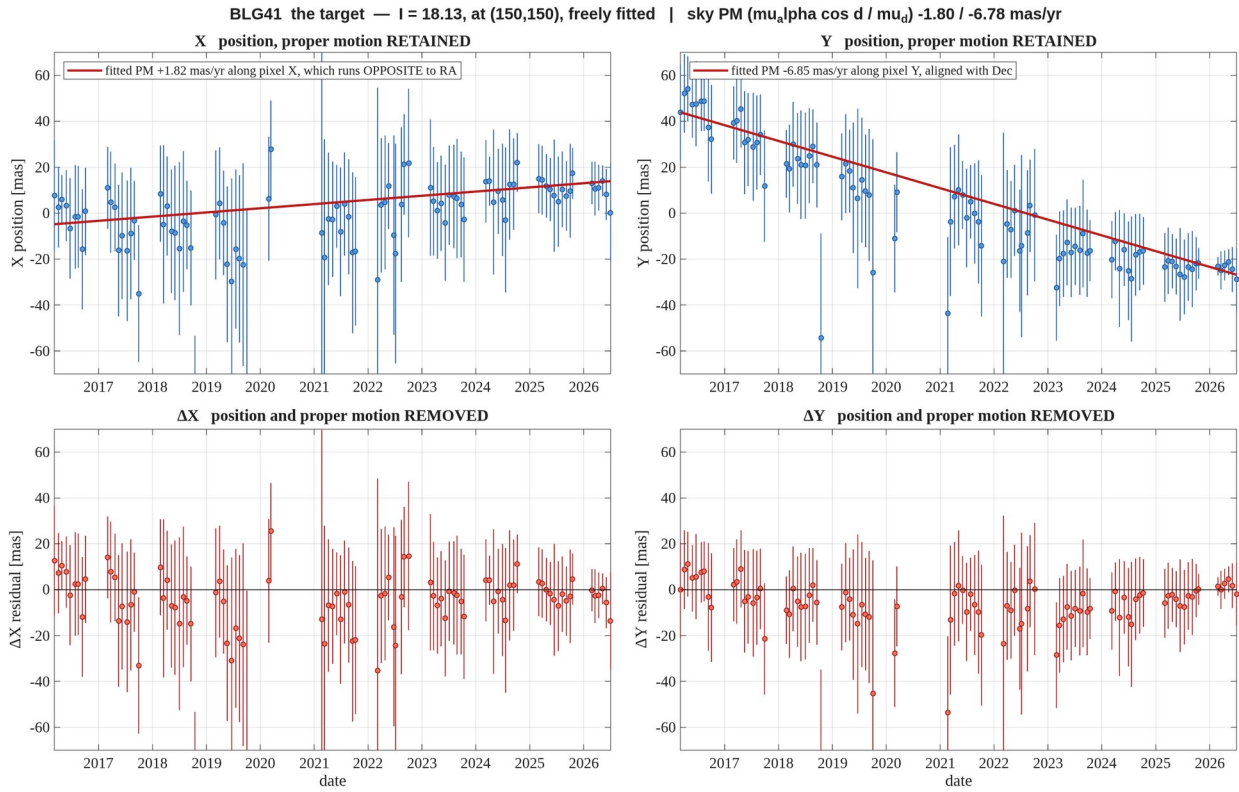


Figure 2. **BLG41** — the target, freely fitted. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.

report\_v4\_motion\_BLG41\_target.png

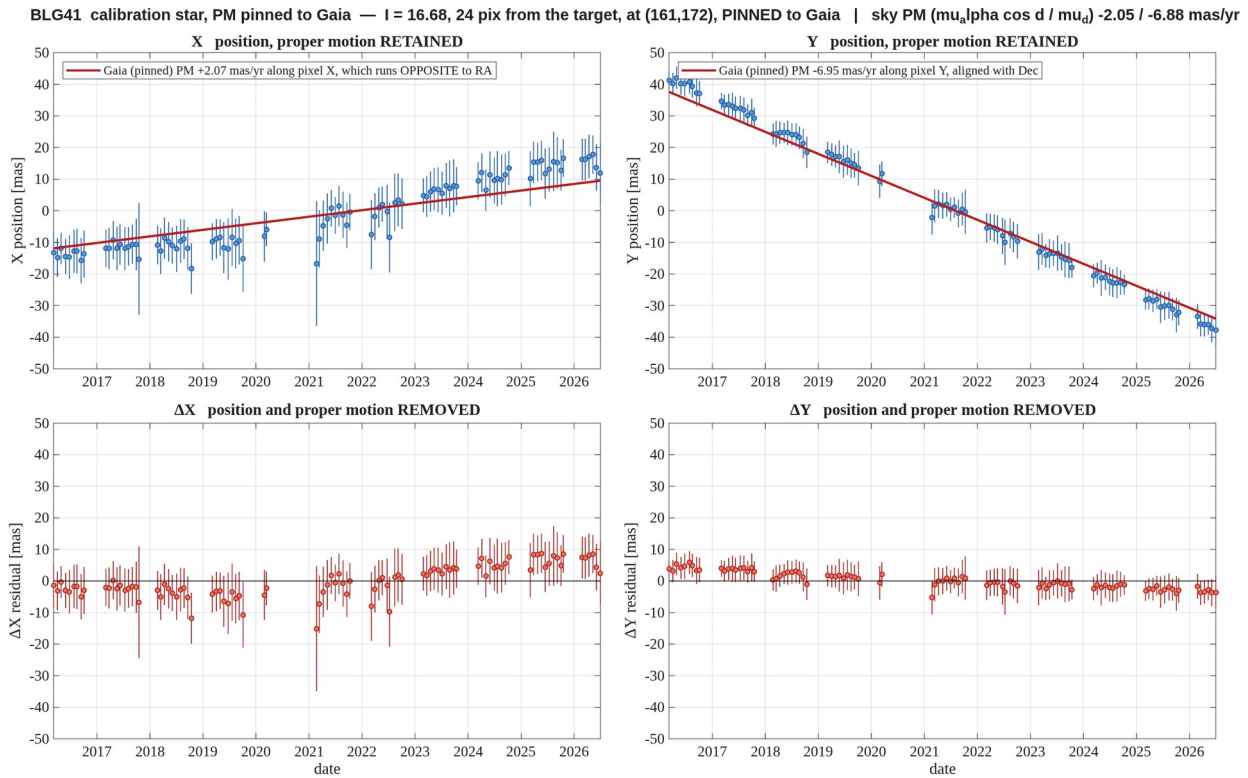


Figure 3. **BLG41** — calibration star 24 pix from the target, proper motion pinned to Gaia. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.  
report\_v4\_motion\_BLG41\_cal\_d24.png

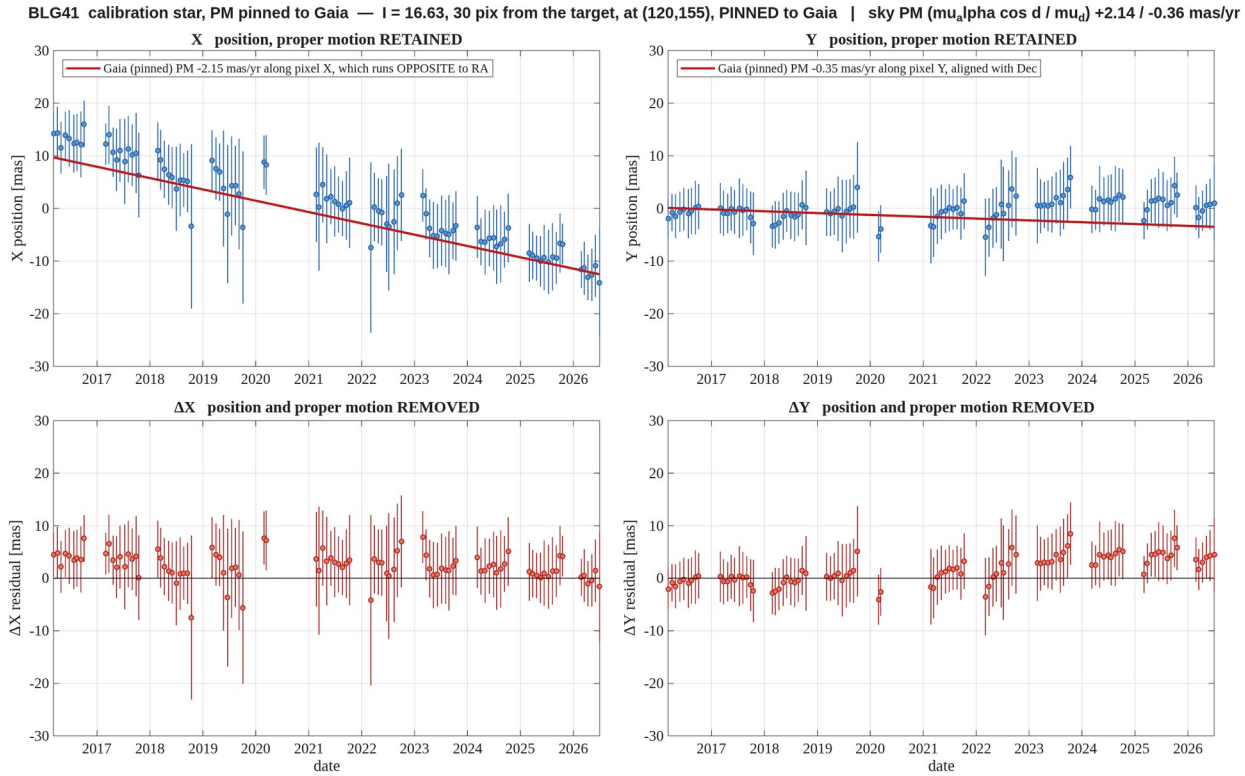


Figure 4. **BLG41** — calibration star 30 pix from the target, proper motion pinned to Gaia. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.  
report\_v4\_motion\_BLG41\_cal\_d30.png

BLG41 calibration star, PM pinned to Gaia —  $I = 16.94$ , 43 pix from the target, at (111,167), PINNED to Gaia | sky PM ( $\mu_\alpha \cos d / \mu_\delta$ )  $+0.23 / -7.85$  mas/yr

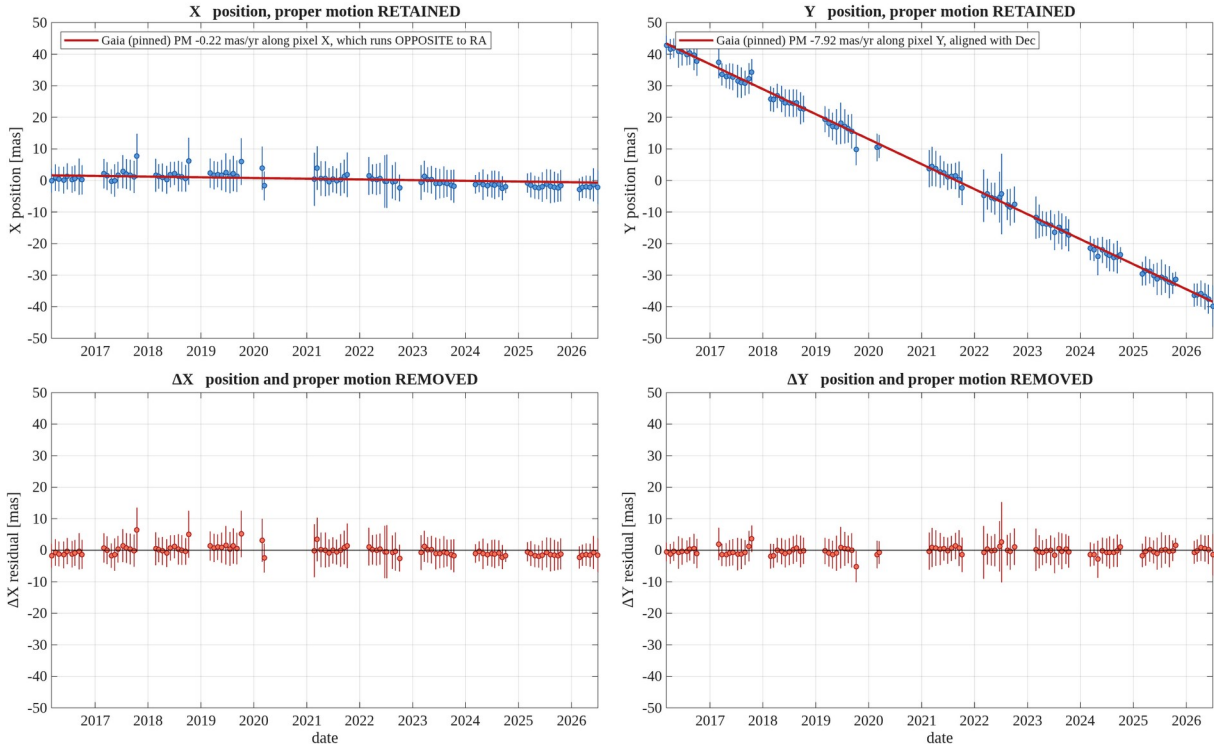


Figure 5. **BLG41** — calibration star 43 pix from the target, proper motion pinned to Gaia. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.

report\_v4\_motion\_BLG41\_cal\_d43.png

BLG41 comparison star outside the set, PM fitted —  $I = 18.14$ , 5 pix from the target, at (146,148), freely fitted | sky PM ( $\mu_\alpha \cos d / \mu_\delta$ )  $-0.63 / -5.47$  mas/yr

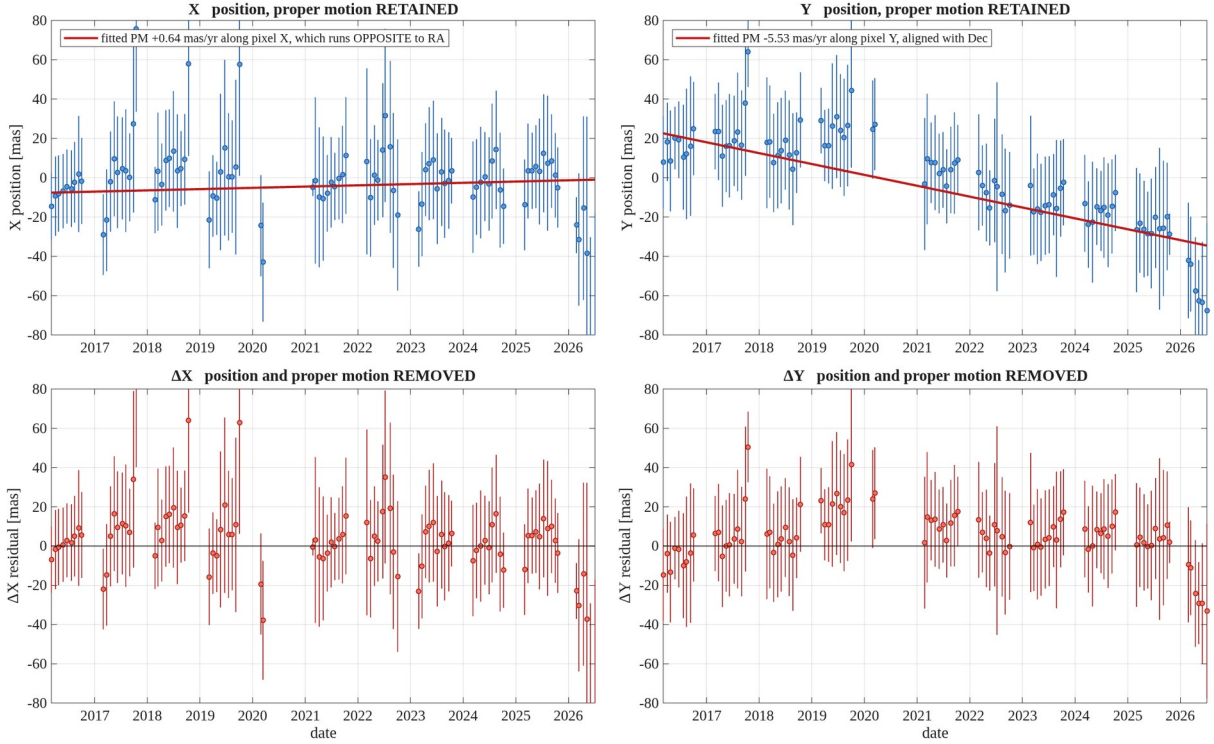




Figure 6. **BLG41** — passenger at the target magnitude, 4.5 pix away — **blended with the target**, see section 7. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.  
report\_v4\_motion\_BLG41\_out\_d04.png

**BLG41 comparison star outside the set, PM fitted** —  $I = 17.91$ , 21 pix from the target, at (155,171), freely fitted | sky PM ( $\mu_\alpha \cos d / \mu_\delta$ ) -6.81 / -9.99 mas/yr

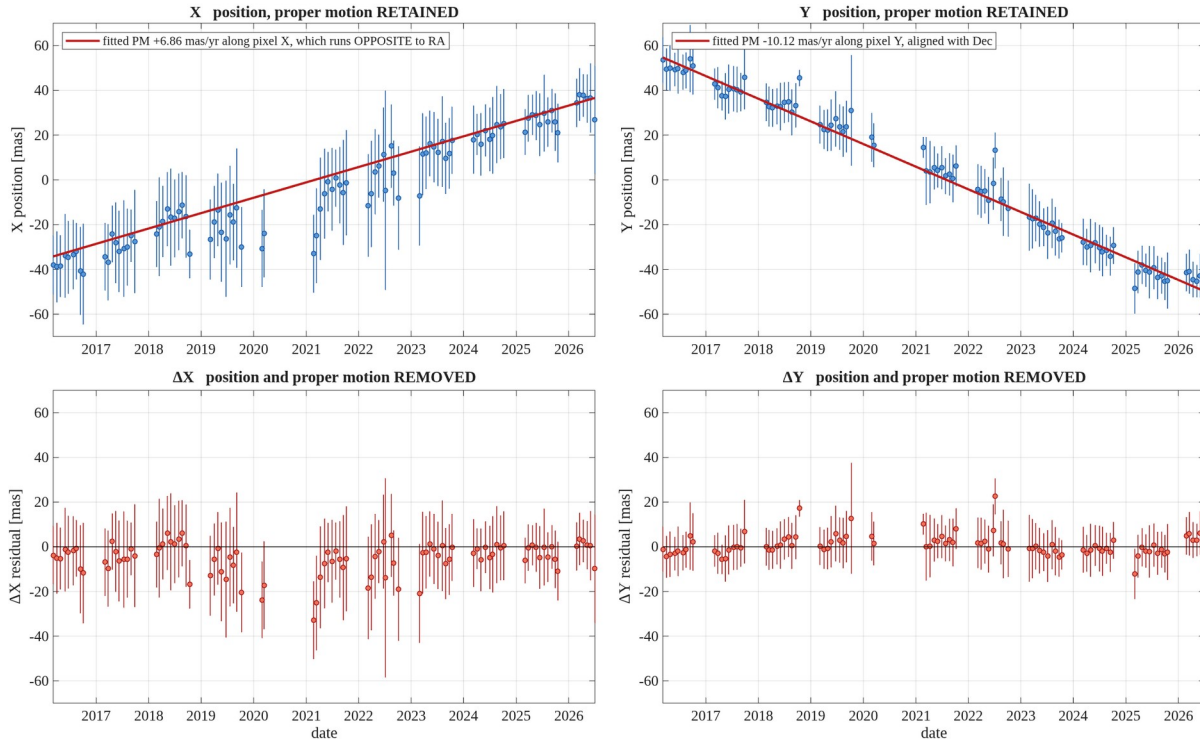


Figure 7. **BLG41** — passenger at the target magnitude, 21 pix away, freely fitted. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.  
report\_v4\_motion\_BLG41\_out\_d21.png

BLG41 comparison star outside the set, PM fitted —  $I = 18.05$ , 48 pix from the target, at (199,145), freely fitted | sky PM ( $\mu_\alpha \cos d / \mu_\delta$ ) +3.06 / -6.34 mas/yr

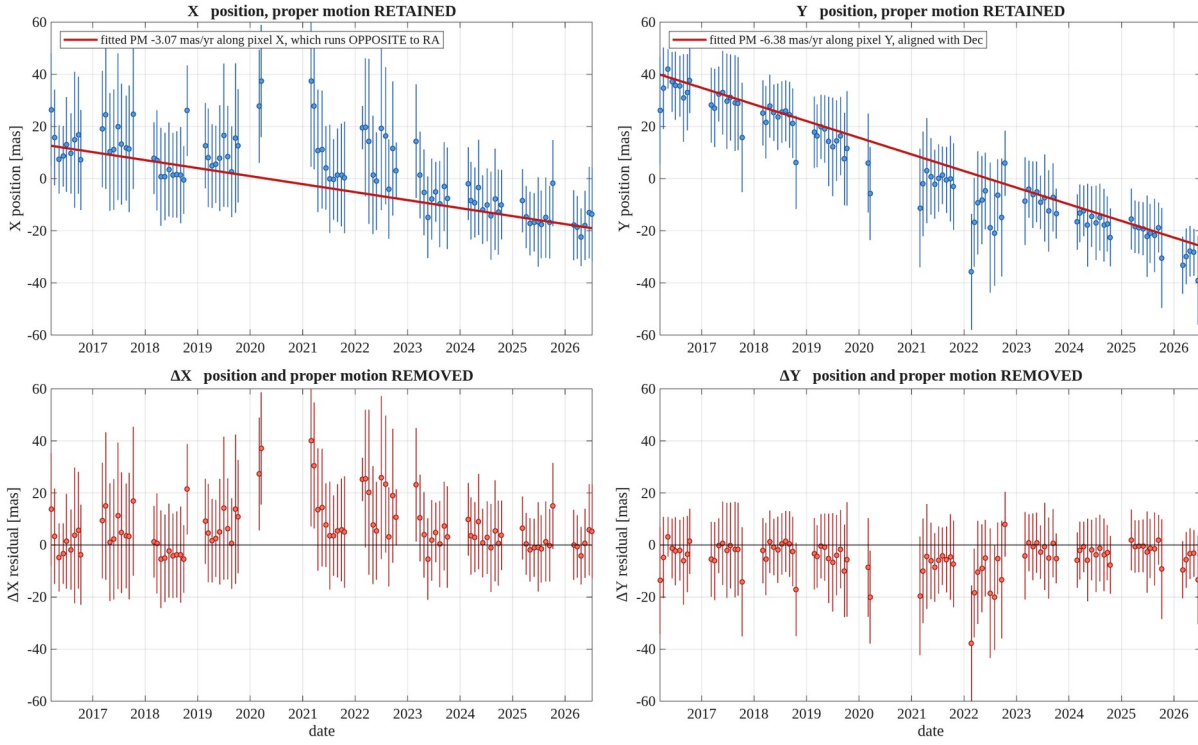


Figure 8. **BLG41** — passenger at the target magnitude, 48 pix away, freely fitted. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.

report\_v4\_motion\_BLG41\_out\_d48.png

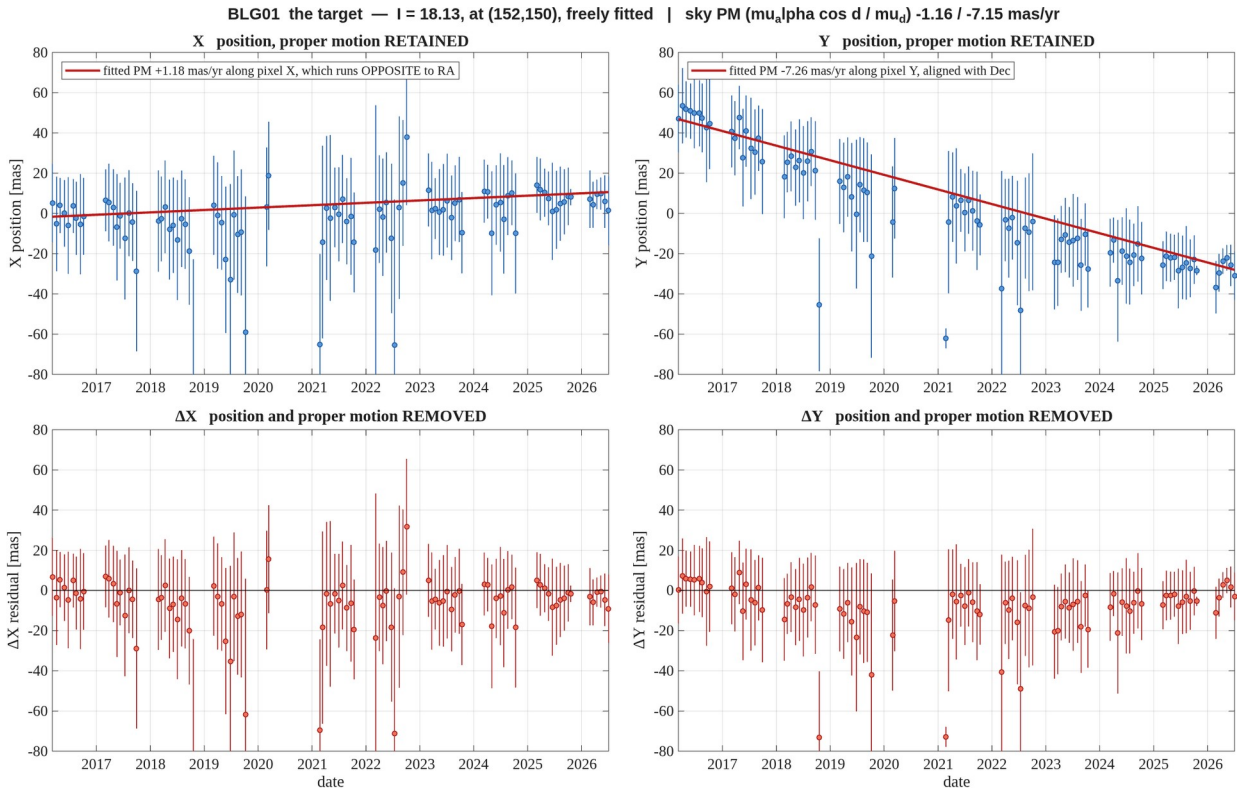




Figure 9. **BLG01** — the target, freely fitted. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.

report\_v4\_motion\_BLG01\_target.png

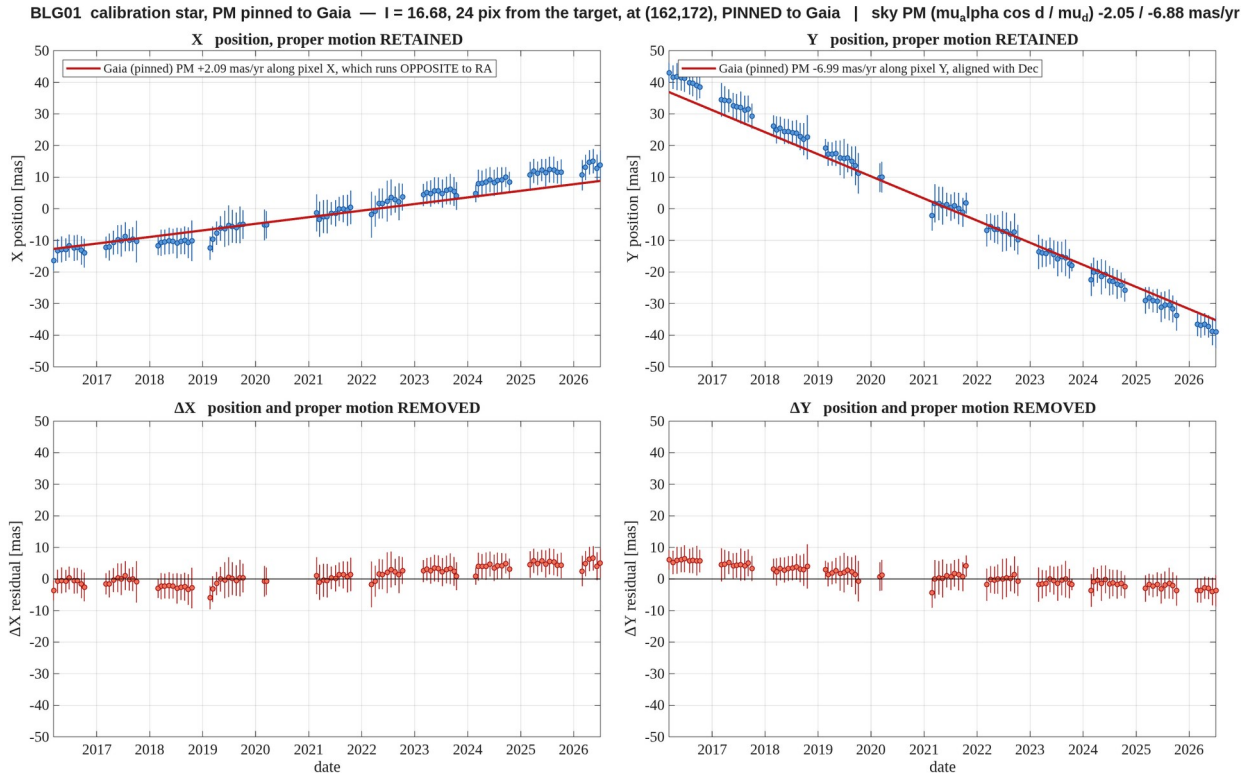


Figure 10. **BLG01** — calibration star 24 pix from the target, proper motion pinned to Gaia. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.

report\_v4\_motion\_BLG01\_cal\_d24.png

BLG01 calibration star, PM pinned to Gaia —  $I = 16.63$ , 30 pix from the target, at (121,154), PINNED to Gaia | sky PM ( $\mu_\alpha \cos d / \mu_d$ )  $+2.14 / -0.36$  mas/yr

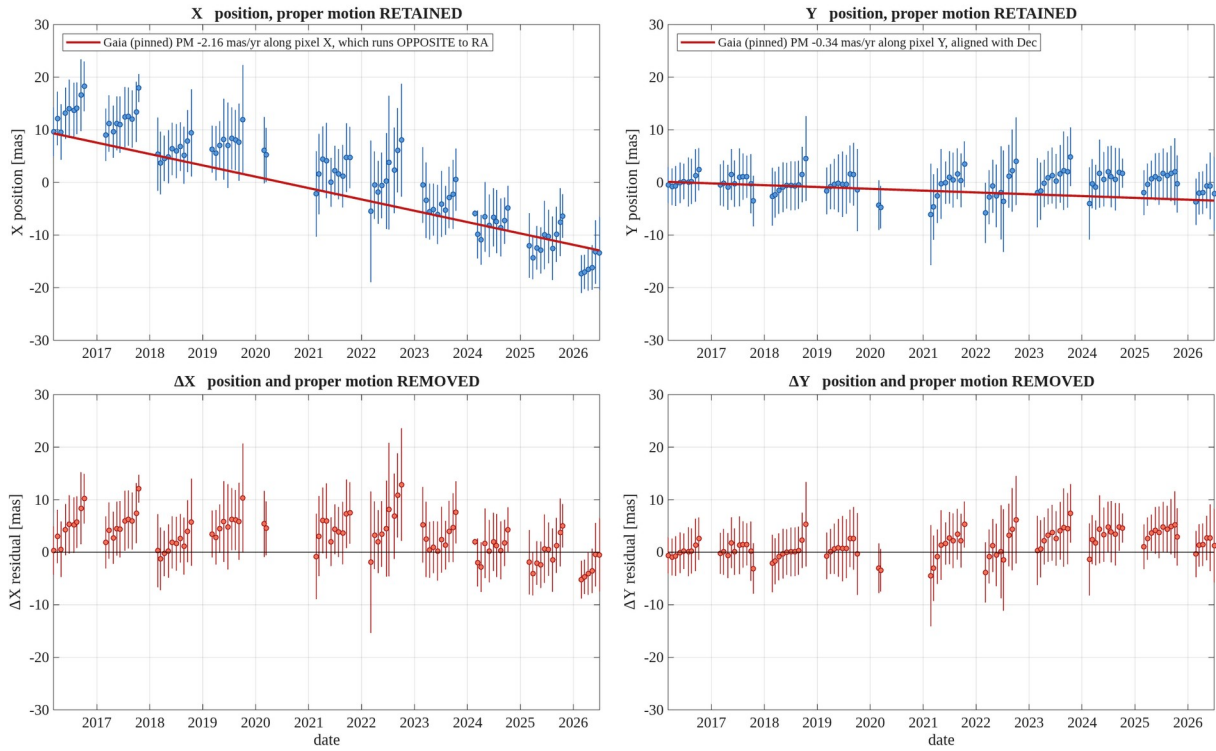


Figure 11. **BLG01** — calibration star 30 pix from the target, proper motion pinned to Gaia. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.

report\_v4\_motion\_BLG01\_cal\_d30.png

BLG01 calibration star, PM pinned to Gaia —  $I = 16.94$ , 43 pix from the target, at (112,166), PINNED to Gaia | sky PM ( $\mu_\alpha \cos d / \mu_d$ )  $+0.23 / -7.85$  mas/yr

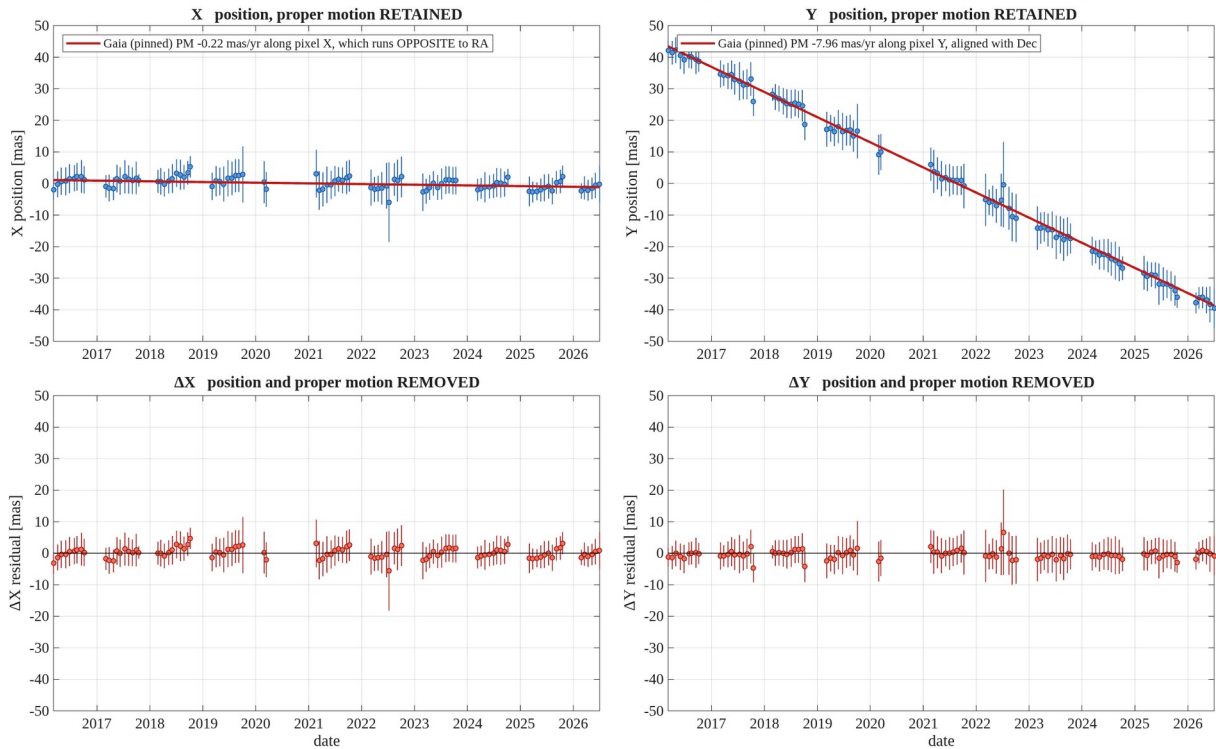


Figure 12. **BLG01** — calibration star 43 pix from the target, proper motion pinned to Gaia. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.  
report\_v4\_motion\_BLG01\_cal\_d43.png

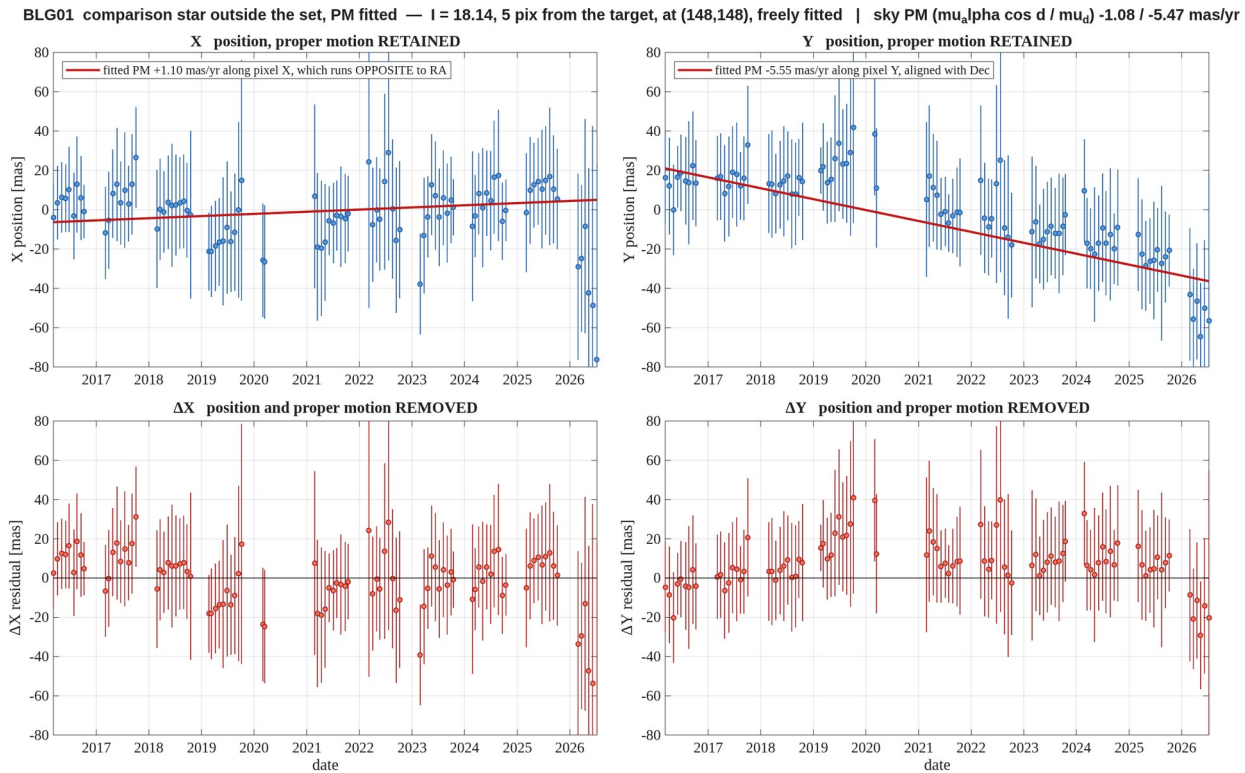


Figure 13. **BLG01** — passenger at the target magnitude, 4.5 pix away — **blended with the target**, see section 7. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.  
report\_v4\_motion\_BLG01\_out\_d04.png

BLG01 comparison star outside the set, PM fitted —  $l = 17.91$ , 21 pix from the target, at (156,171), freely fitted | sky PM ( $\mu_\alpha \cos \delta / \mu_\delta$ ) -6.94 / -10.06 mas/yr

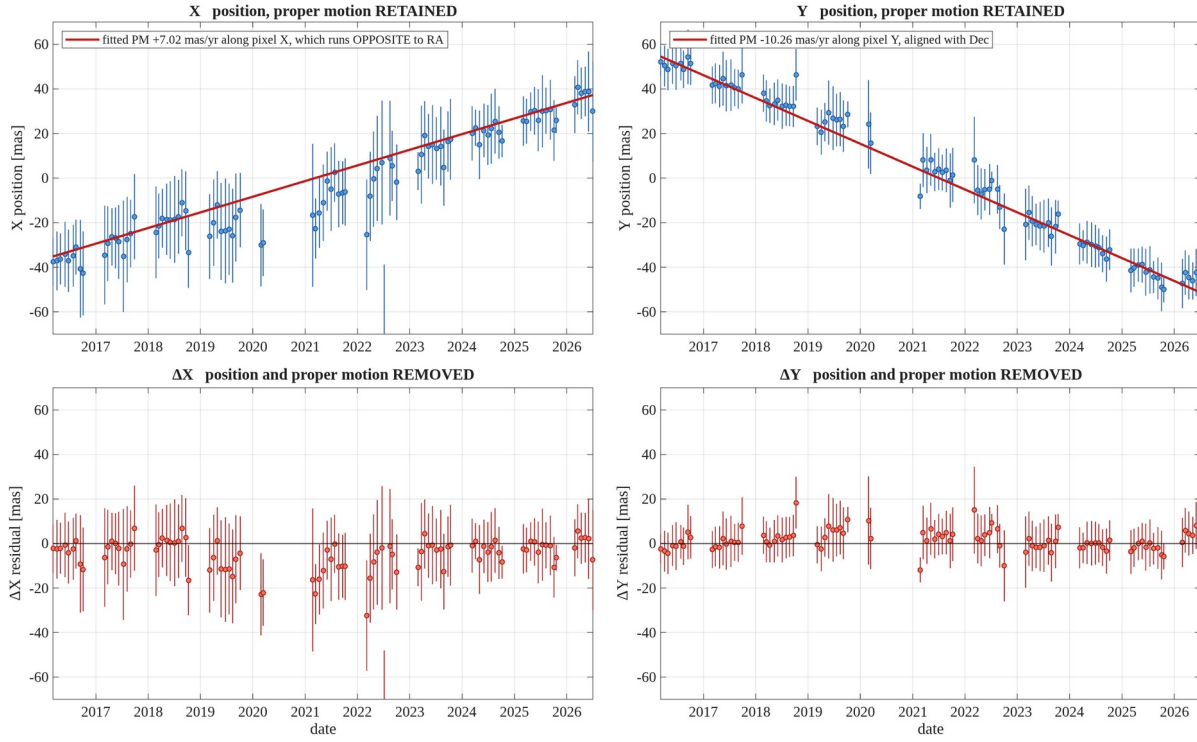


Figure 14. **BLG01** — passenger at the target magnitude, 21 pix away, freely fitted. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.  
report\_v4\_motion\_BLG01\_out\_d21.png

BLG01 comparison star outside the set, PM fitted —  $l = 18.05$ , 49 pix from the target, at (200,145), freely fitted | sky PM ( $\mu_\alpha \cos \delta / \mu_\delta$ ) +2.80 / -6.48 mas/yr

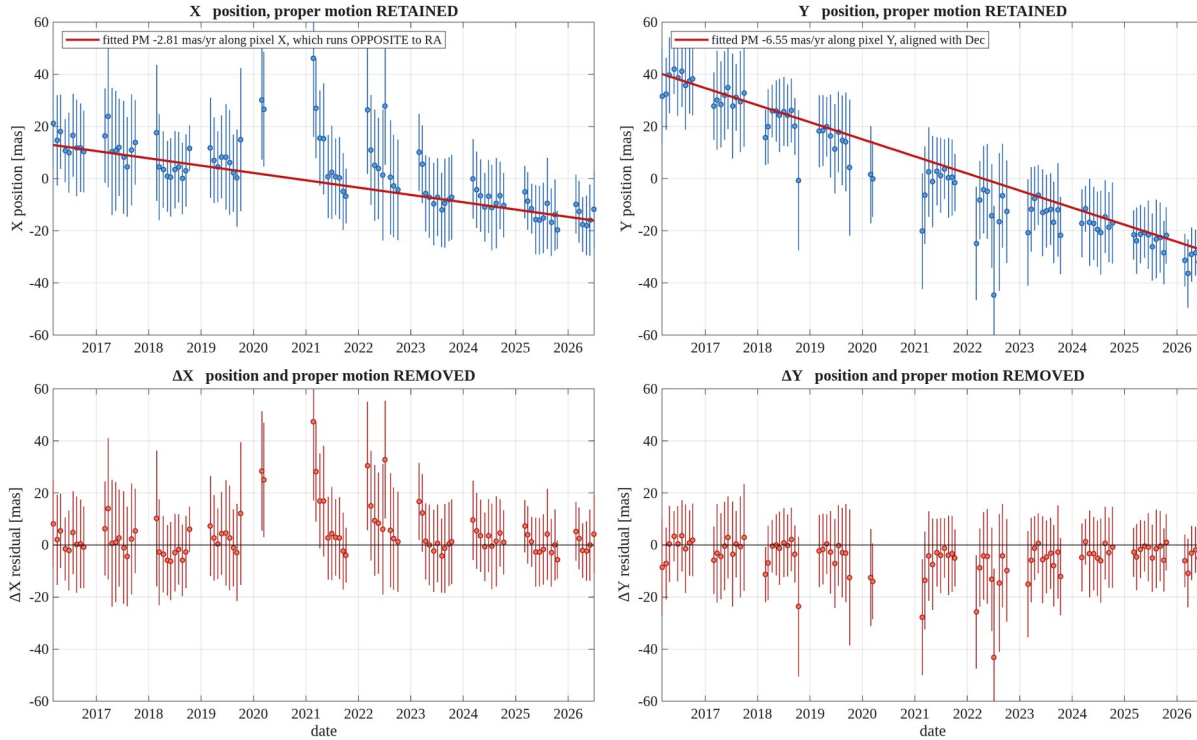


Figure 15. **BLG01** — passenger at the target magnitude, 48 pix away, freely fitted. Binned in one sidereal month; error bars are the RMS within the bin. Columns are the two axes, **X left and Y right**; the top row keeps the proper motion, the bottom row removes it. The heading gives the star's magnitude, its distance from the target and its sky proper motion. **A pinned star's red line is Gaia's proper motion, not a fitted one.** The legend quotes the motion along the pixel axis and the heading quotes it on the sky; **pixel X runs opposite to RA**, so those two carry opposite signs in X.  
report\_v4\_motion\_BLG01\_out\_d48.png



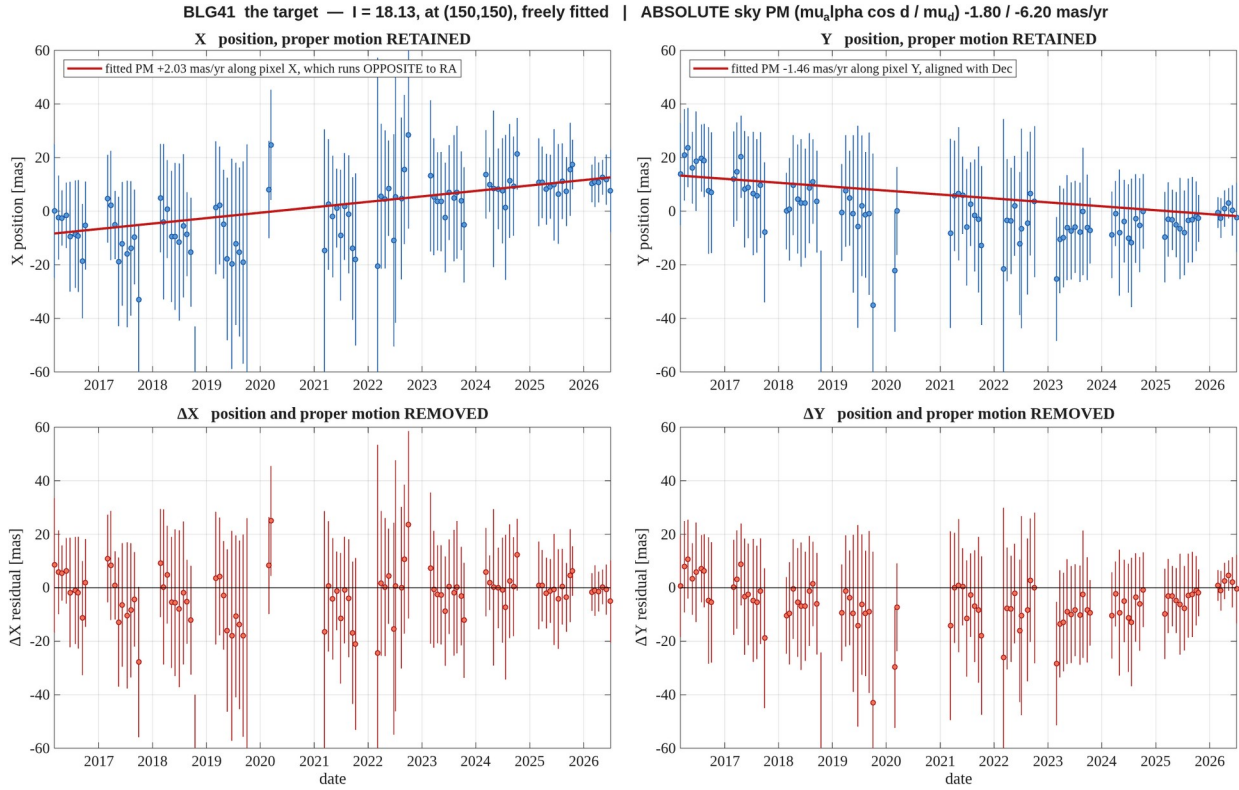


Figure 16. **v5, BLG41** — the target, freely fitted. The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X.

report\_v5\_motion\_BLG41\_target.png

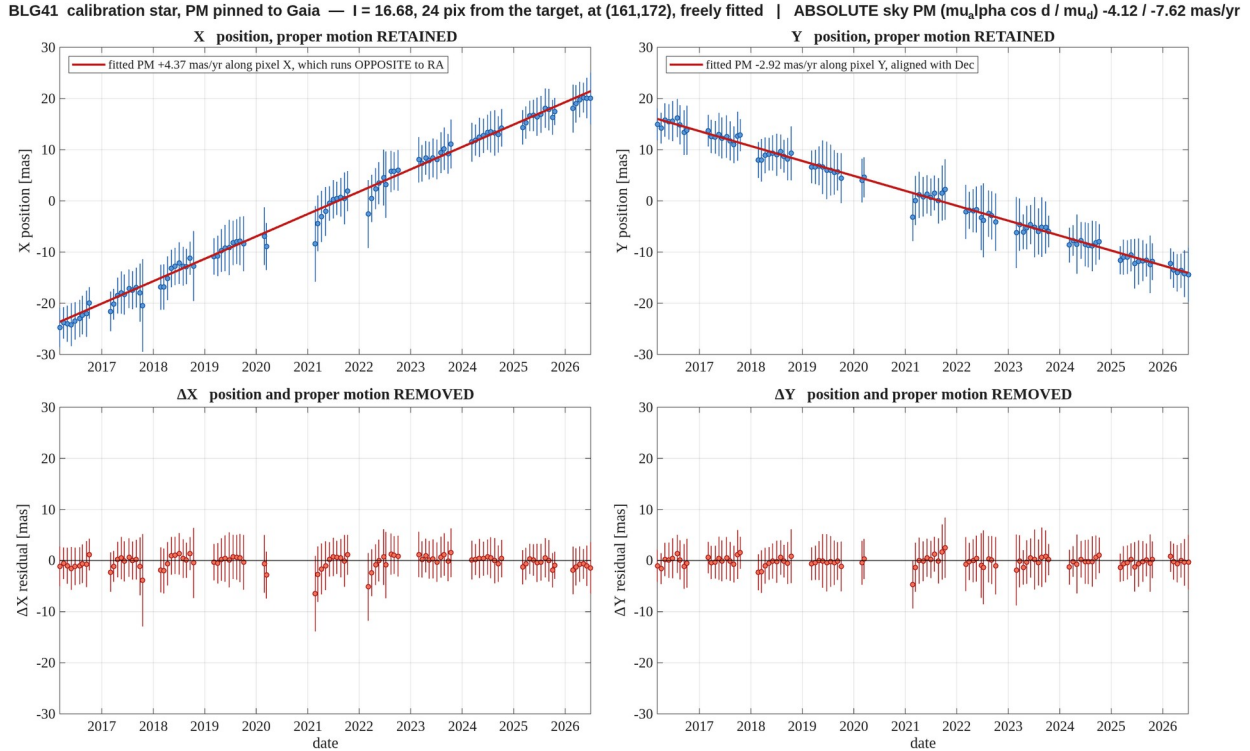


Figure 17. **v5, BLG41** — calibration star 24 pix from the target, freely fitted (v5 pins nothing). The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X.

report\_v5\_motion\_BLG41\_cal\_d24.png

BLG41 calibration star, PM pinned to Gaia —  $l = 16.63$ , 30 pix from the target, at (120,155), freely fitted | ABSOLUTE sky PM ( $\mu_\alpha \cos d / \mu_\delta$ )  $+2.20 / +1.95$  mas/yr

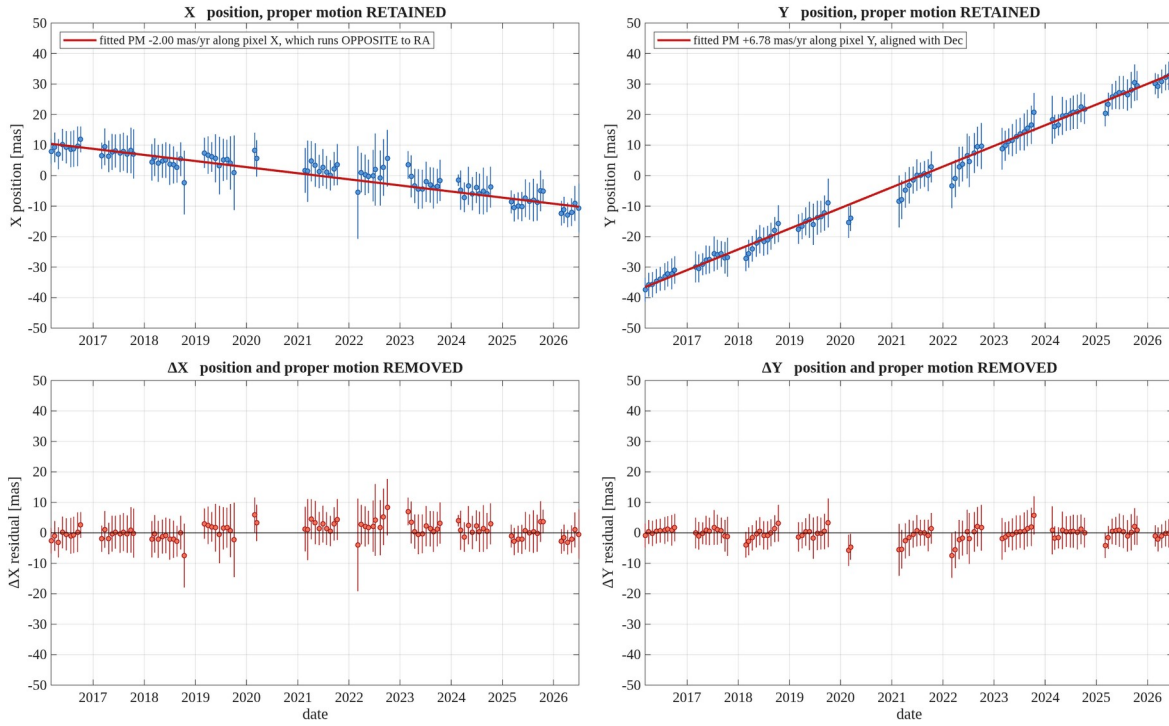


Figure 18. **v5, BLG41** — calibration star 30 pix from the target, freely fitted (v5 pins nothing). The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X. `report_v5_motion_BLG41_cal_d30.png`

BLG41 calibration star, PM pinned to Gaia —  $l = 16.94$ , 43 pix from the target, at (111,167), freely fitted | ABSOLUTE sky PM ( $\mu_\alpha \cos d / \mu_\delta$ )  $-0.28 / -5.96$  mas/yr

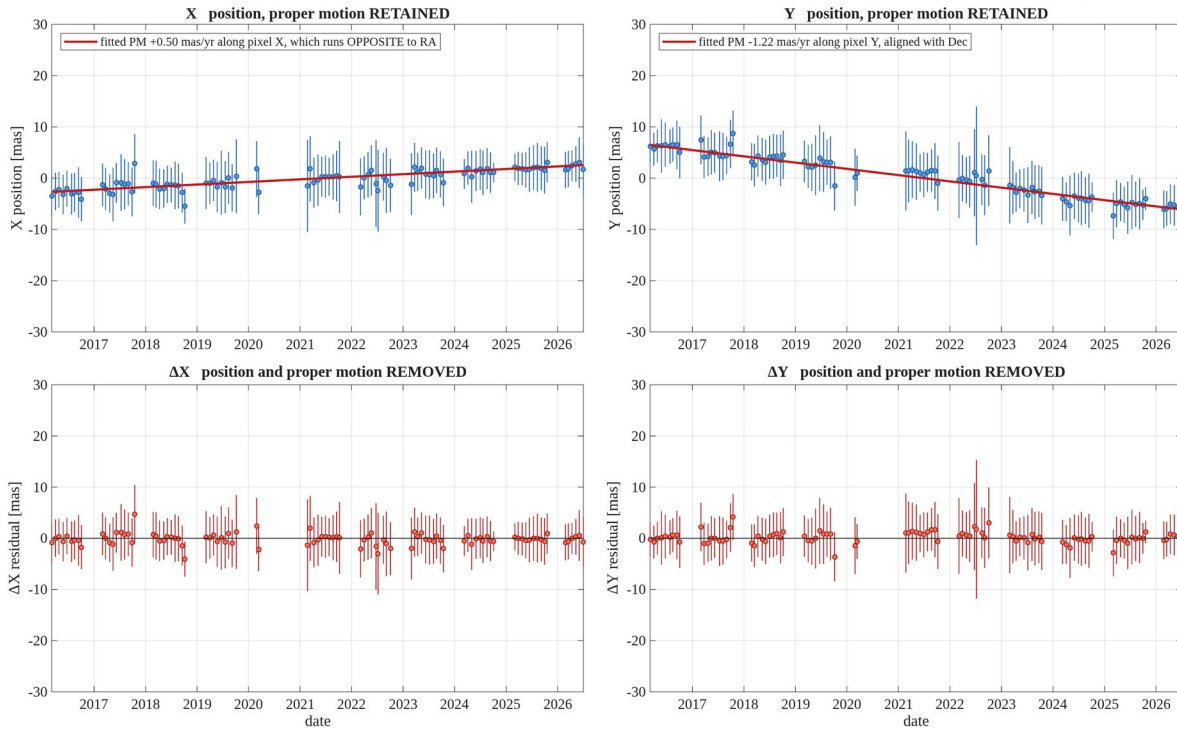


Figure 19. **v5, BLG41** — calibration star 43 pix from the target, freely fitted (v5 pins nothing). The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X. `report_v5_motion_BLG41_cal_d43.png`

BLG41 comparison star outside the set, PM fitted —  $l = 18.14$ , 5 pix from the target, at (146,148), freely fitted | ABSOLUTE sky PM ( $\mu_\alpha \cos \delta / \mu_\delta$ )  $-0.17 / -4.55$  mas/y

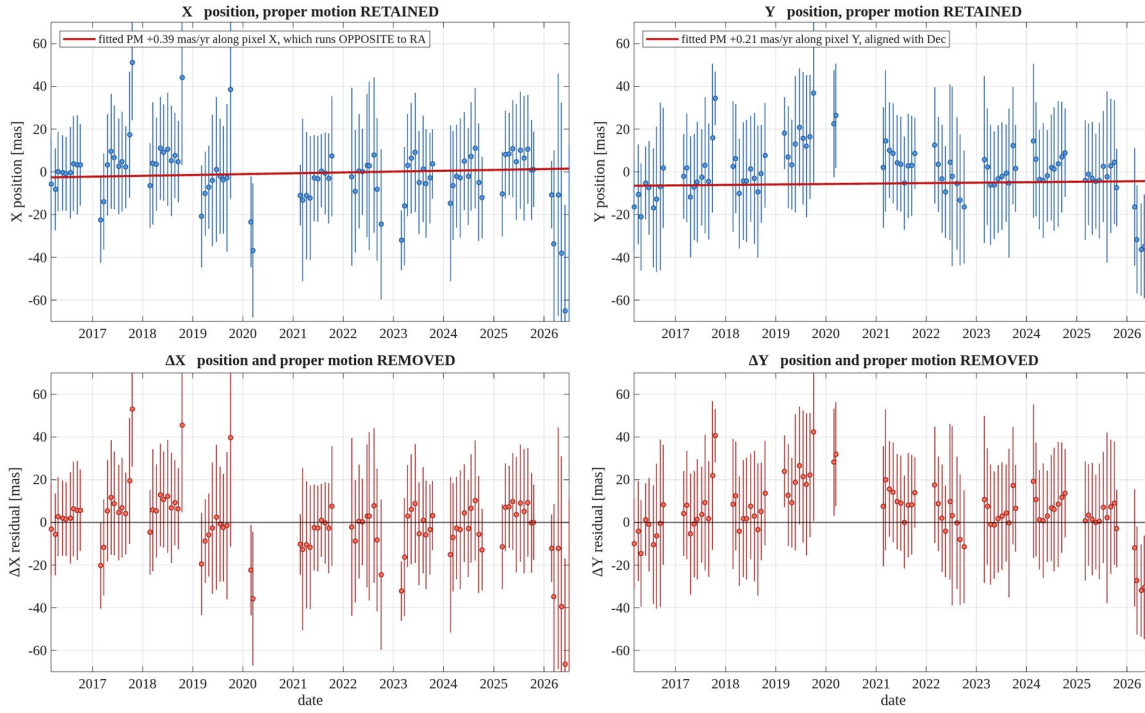


Figure 20. **v5, BLG41** — passenger at the target magnitude, 4.5 pix away — **blended with the target**, see section 7. The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X.

report\_v5\_motion\_BLG41\_out\_d04.png

BLG41 comparison star outside the set, PM fitted —  $l = 17.91$ , 21 pix from the target, at (155,171), freely fitted | ABSOLUTE sky PM ( $\mu_\alpha \cos \delta / \mu_\delta$ )  $-7.71 / -9.55$  mas/y

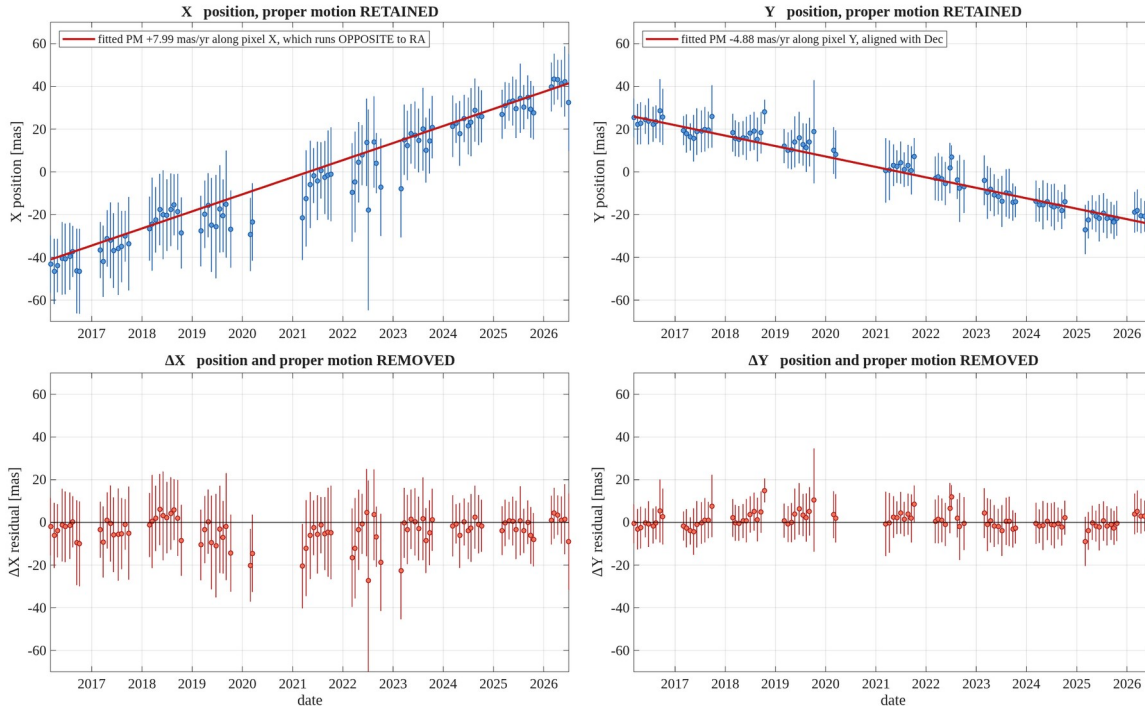


Figure 21. **v5, BLG41** — passenger at the target magnitude, 21 pix away, freely fitted. The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X.

report\_v5\_motion\_BLG41\_out\_d21.png

LG41 comparison star outside the set, PM fitted —  $l = 18.05$ , 48 pix from the target, at (199,145), freely fitted | ABSOLUTE sky PM ( $\mu_\alpha \cos d / \mu_\delta$ ) +3.08 / -7.56 mas/

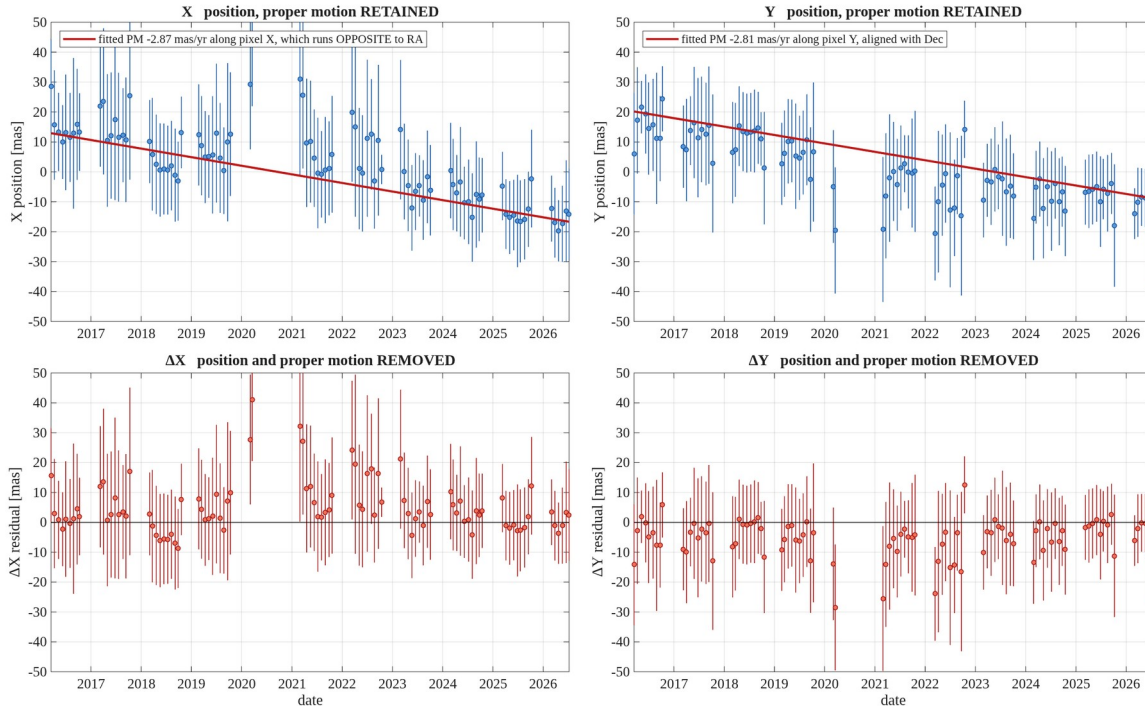


Figure 22. **v5, BLG41** — passenger at the target magnitude, 48 pix away, freely fitted. The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X. report\_v5\_motion\_BLG41\_out\_d48.png

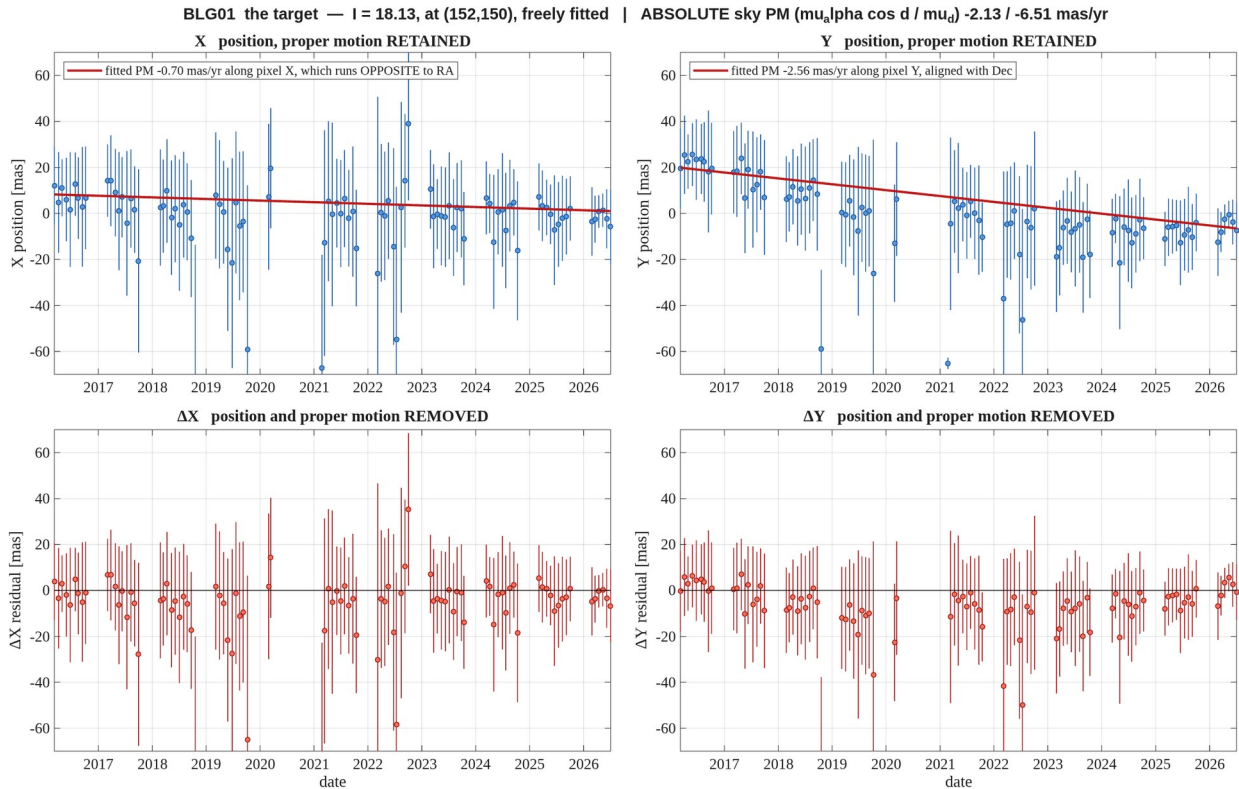


Figure 23. **v5, BLG01** — the target, freely fitted. The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X. report\_v5\_motion\_BLG01\_target.png



BLG01 calibration star, PM pinned to Gaia —  $l = 16.68$ , 24 pix from the target, at (162,172), freely fitted | ABSOLUTE sky PM ( $\mu_\alpha \cos d / \mu_\delta$ ) -4.59 / -7.73 mas/yr

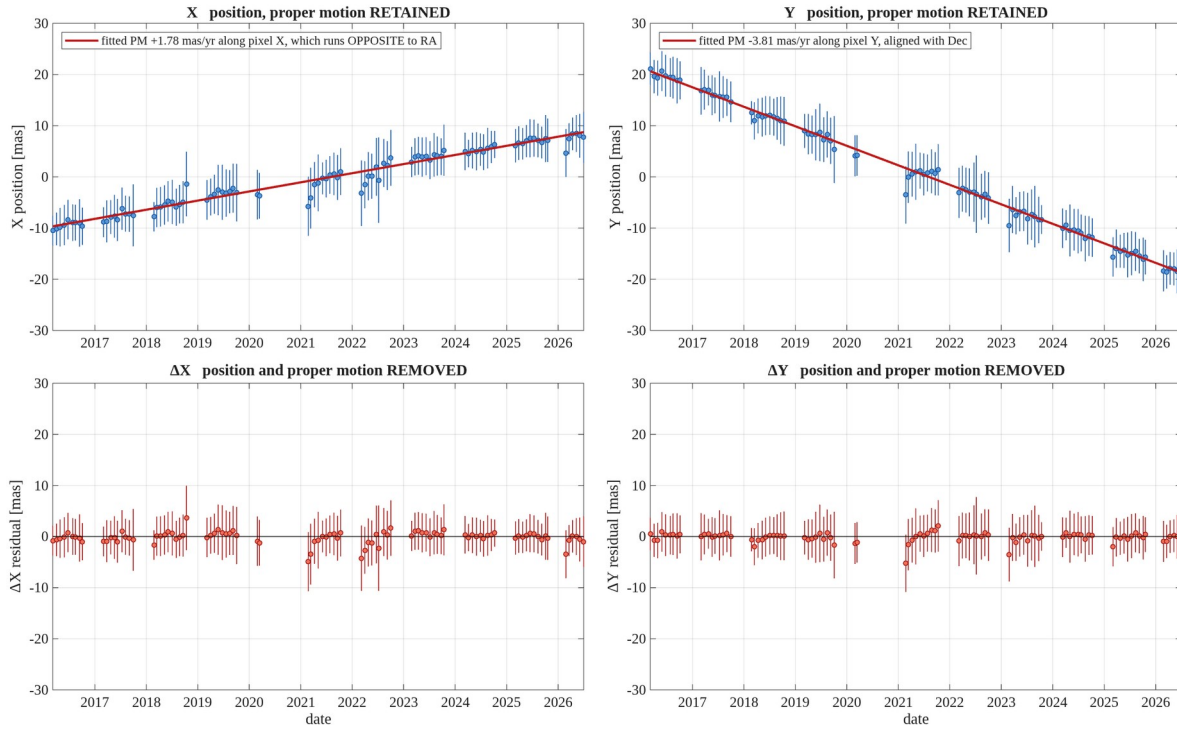


Figure 24. **v5, BLG01** — calibration star 24 pix from the target, freely fitted (v5 pins nothing). The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X.   
report\_v5\_motion\_BLG01\_cal\_d24.png

BLG01 calibration star, PM pinned to Gaia —  $l = 16.63$ , 30 pix from the target, at (121,154), freely fitted | ABSOLUTE sky PM ( $\mu_\alpha \cos d / \mu_\delta$ ) +1.58 / +1.97 mas/yr

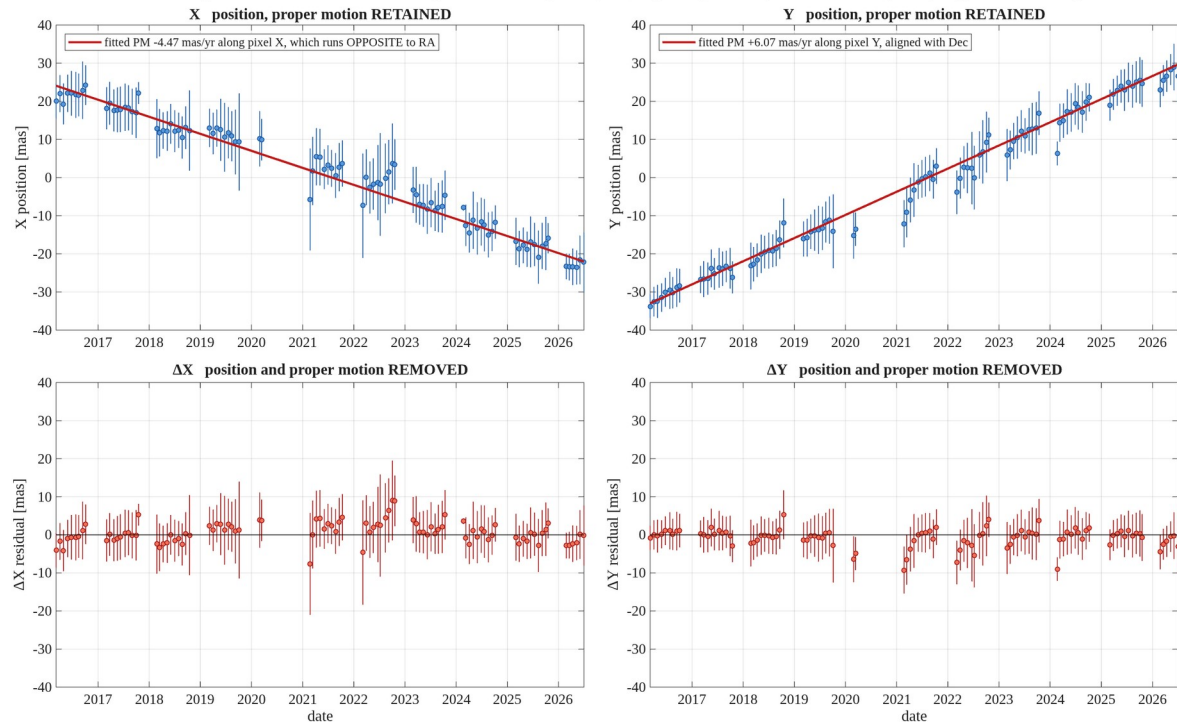


Figure 25. **v5, BLG01** — calibration star 30 pix from the target, freely fitted (v5 pins nothing). The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X.   
report\_v5\_motion\_BLG01\_cal\_d30.png

BLG01 calibration star, PM pinned to Gaia —  $l = 16.94$ , 43 pix from the target, at (112,166), freely fitted | ABSOLUTE sky PM ( $\mu_\alpha \cos d / \mu_\delta$ ) -1.00 / -5.99 mas/yr

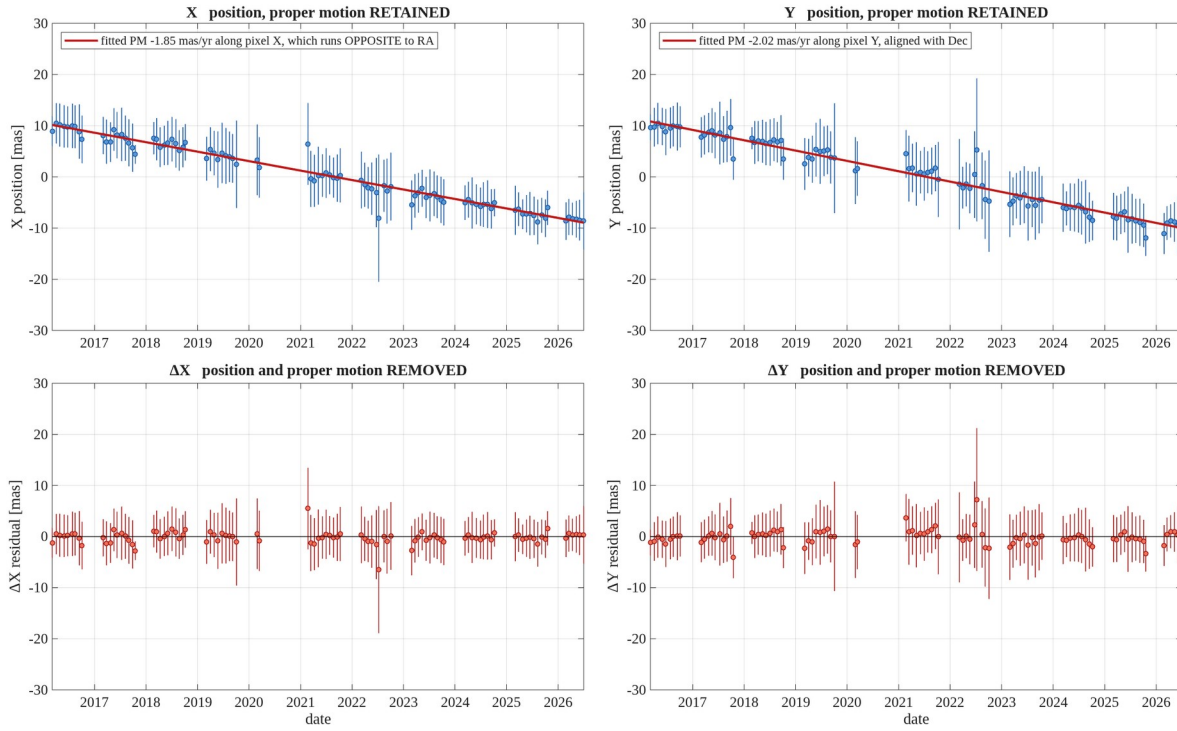


Figure 26. **v5, BLG01** — calibration star 43 pix from the target, freely fitted (v5 pins nothing). The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X.   
report\_v5\_motion\_BLG01\_cal\_d43.png

BLG01 comparison star outside the set, PM fitted —  $l = 18.14$ , 5 pix from the target, at (148,148), freely fitted | ABSOLUTE sky PM ( $\mu_\alpha \cos d / \mu_\delta$ ) -0.71 / -5.42 mas/yr

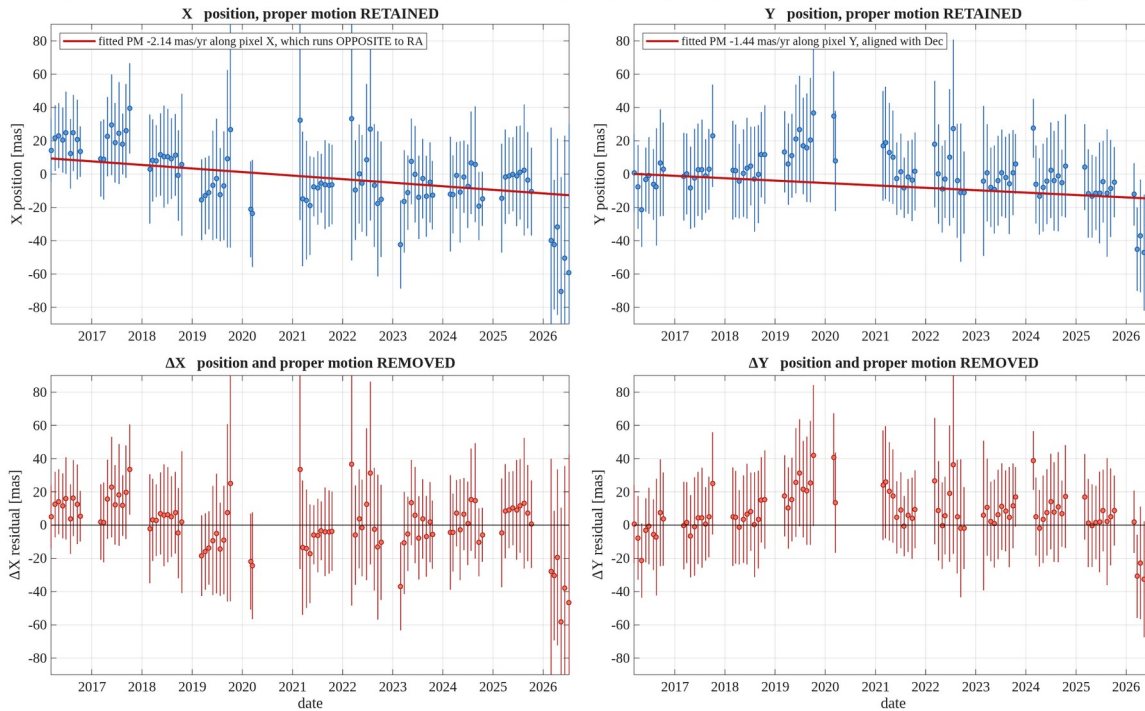


Figure 27. **v5, BLG01** — passenger at the target magnitude, 4.5 pix away — **blended with the target**, see section 7. The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X.   
report\_v5\_motion\_BLG01\_out\_d04.png

LG01 comparison star outside the set, PM fitted —  $l = 17.91$ , 21 pix from the target, at (156,171), freely fitted | ABSOLUTE sky PM ( $\mu_\alpha \cos \delta / \mu_\delta$ ) -8.49 / -9.74 mas/

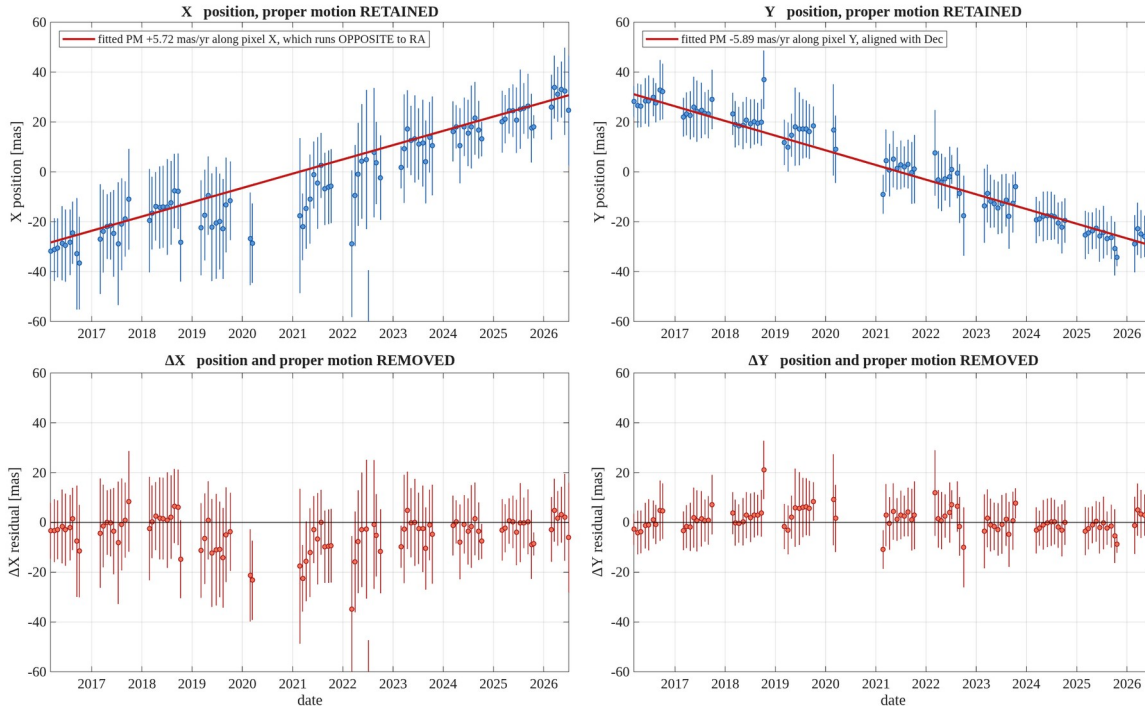


Figure 28. **v5, BLG01** — passenger at the target magnitude, 21 pix away, freely fitted. The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X. report\_v5\_motion\_BLG01\_out\_d21.png

LG01 comparison star outside the set, PM fitted —  $l = 18.05$ , 49 pix from the target, at (200,145), freely fitted | ABSOLUTE sky PM ( $\mu_\alpha \cos \delta / \mu_\delta$ ) +2.83 / -7.32 mas/

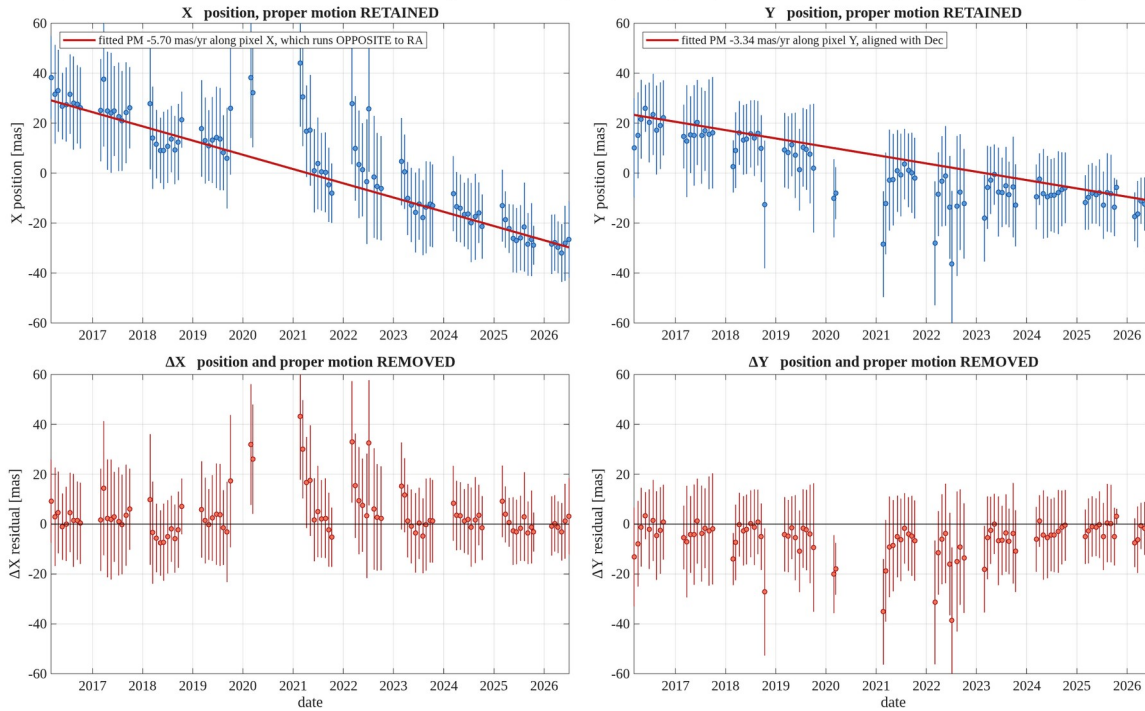


Figure 29. **v5, BLG01** — passenger at the target magnitude, 48 pix away, freely fitted. The same star as the corresponding v4 figure, but from the run in which **nothing is pinned** and Gaia enters only as a post-hoc tie. Sky motions quoted in the heading have the gauge removed and are absolute. Binned in one sidereal month; columns are the two axes, X left and Y right; top row keeps the proper motion, bottom row removes it. **Pixel X runs opposite to RA**, so the legend and the heading carry opposite signs in X. report\_v5\_motion\_BLG01\_out\_d48.png